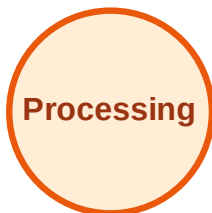


BFS Traversal

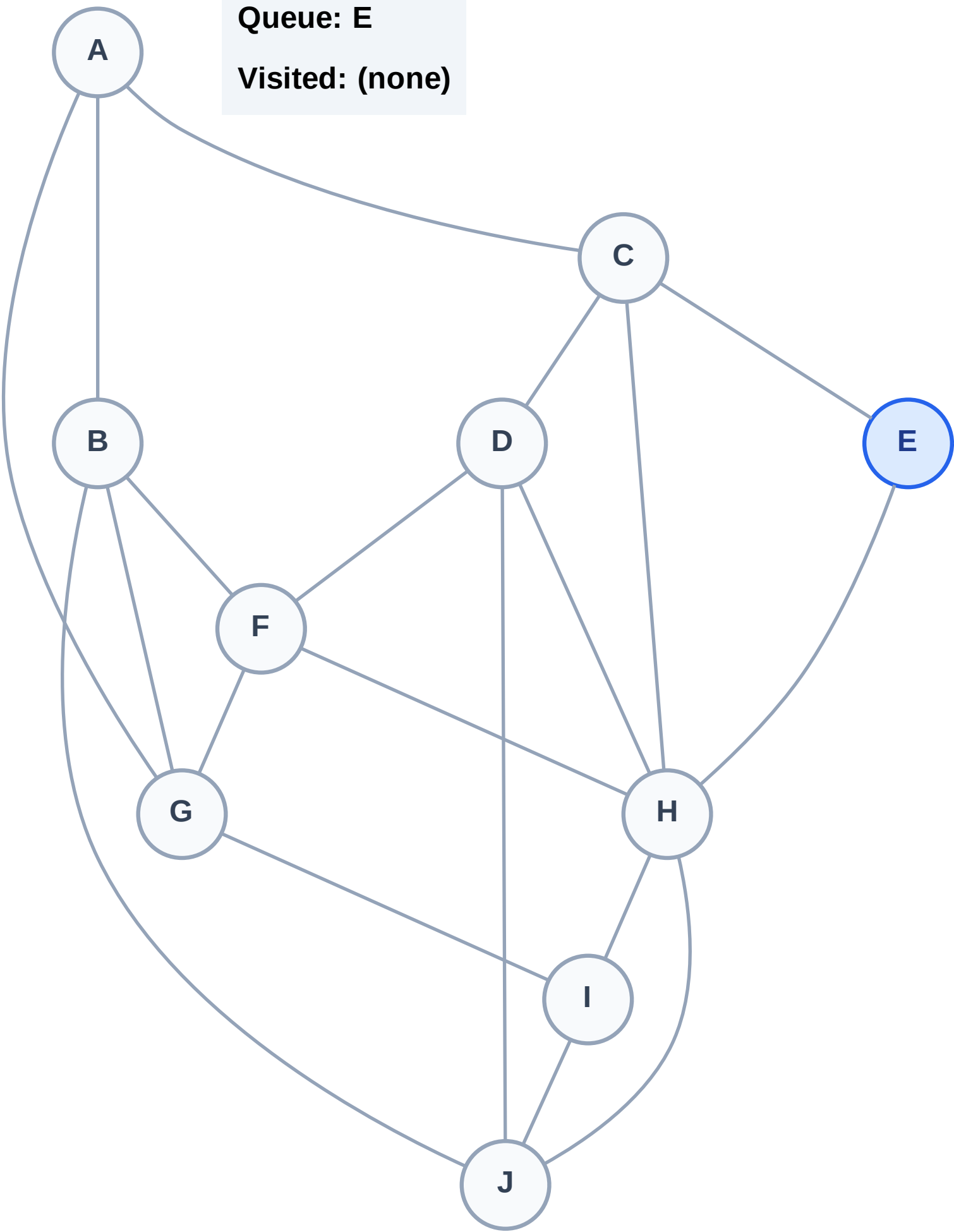
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

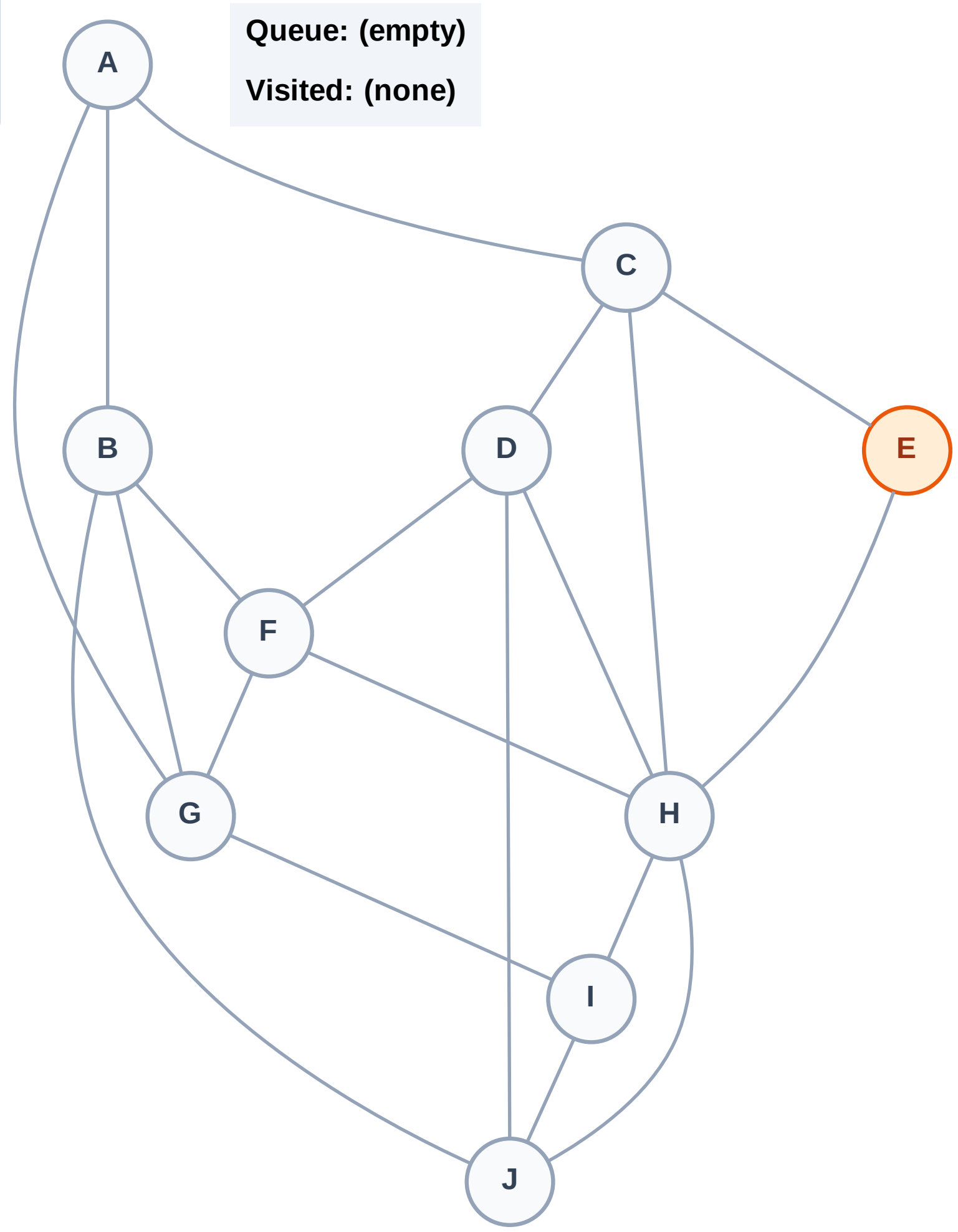
Legend

Complete

Processing

Discovered

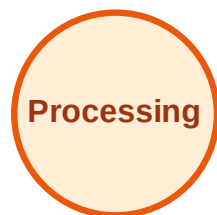
Unseen



BFS Traversal

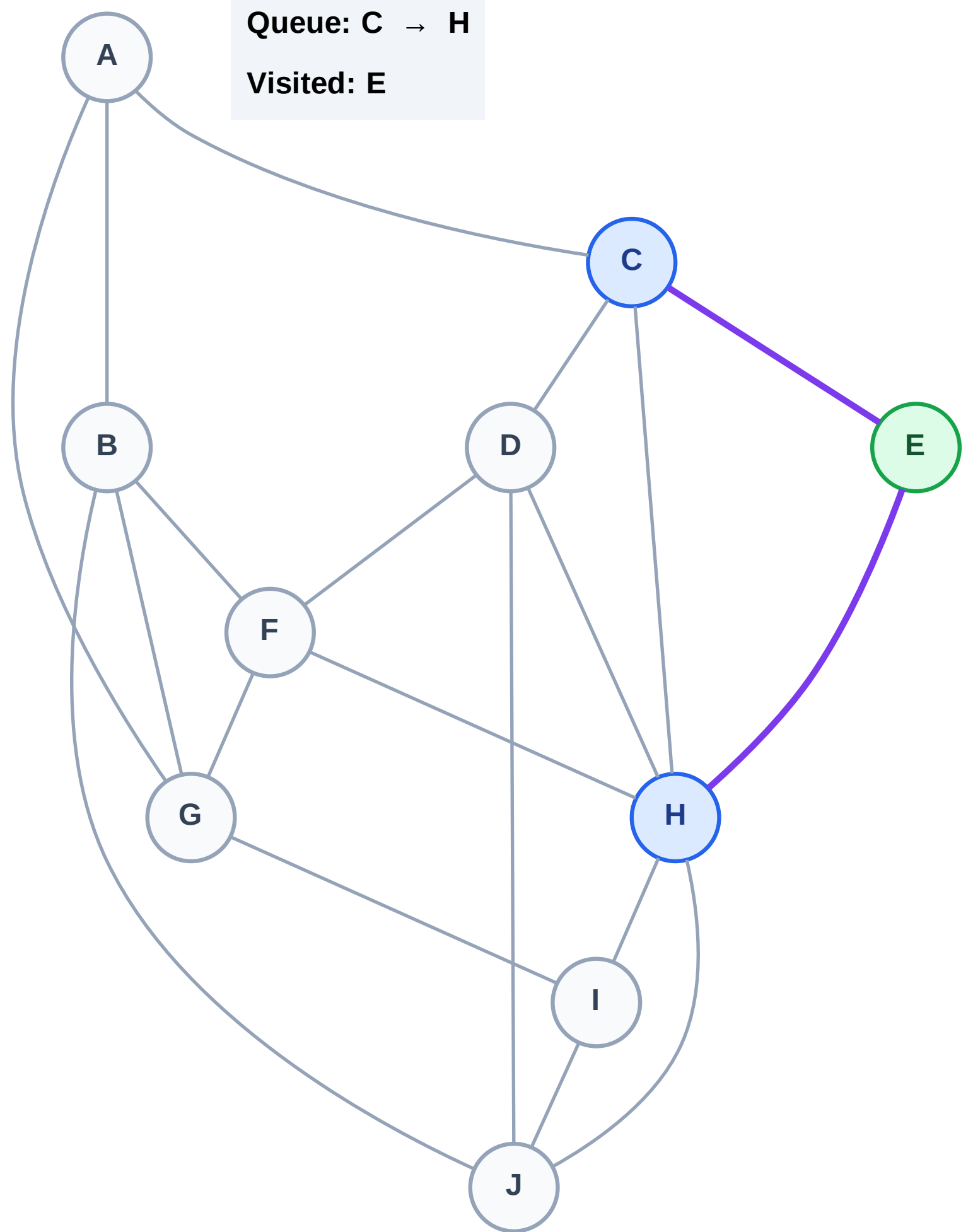
Step 3: Discover C, H from E

Legend



Queue: C → H

Visited: E



BFS Traversal

Step 4: Process node C

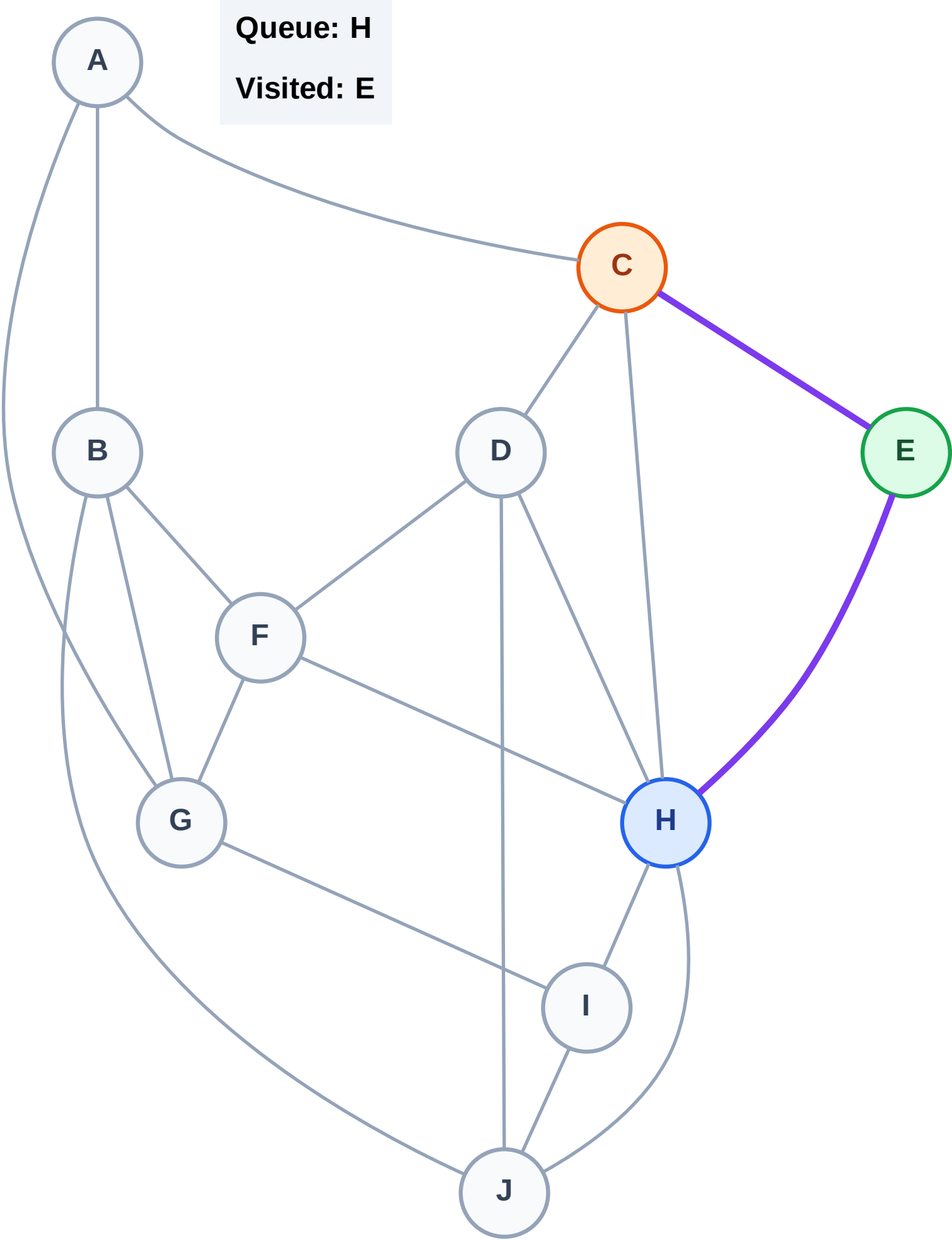
Complete

Processing

Discovered

Unseen

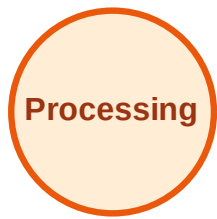
Queue: H
Visited: E



BFS Traversal

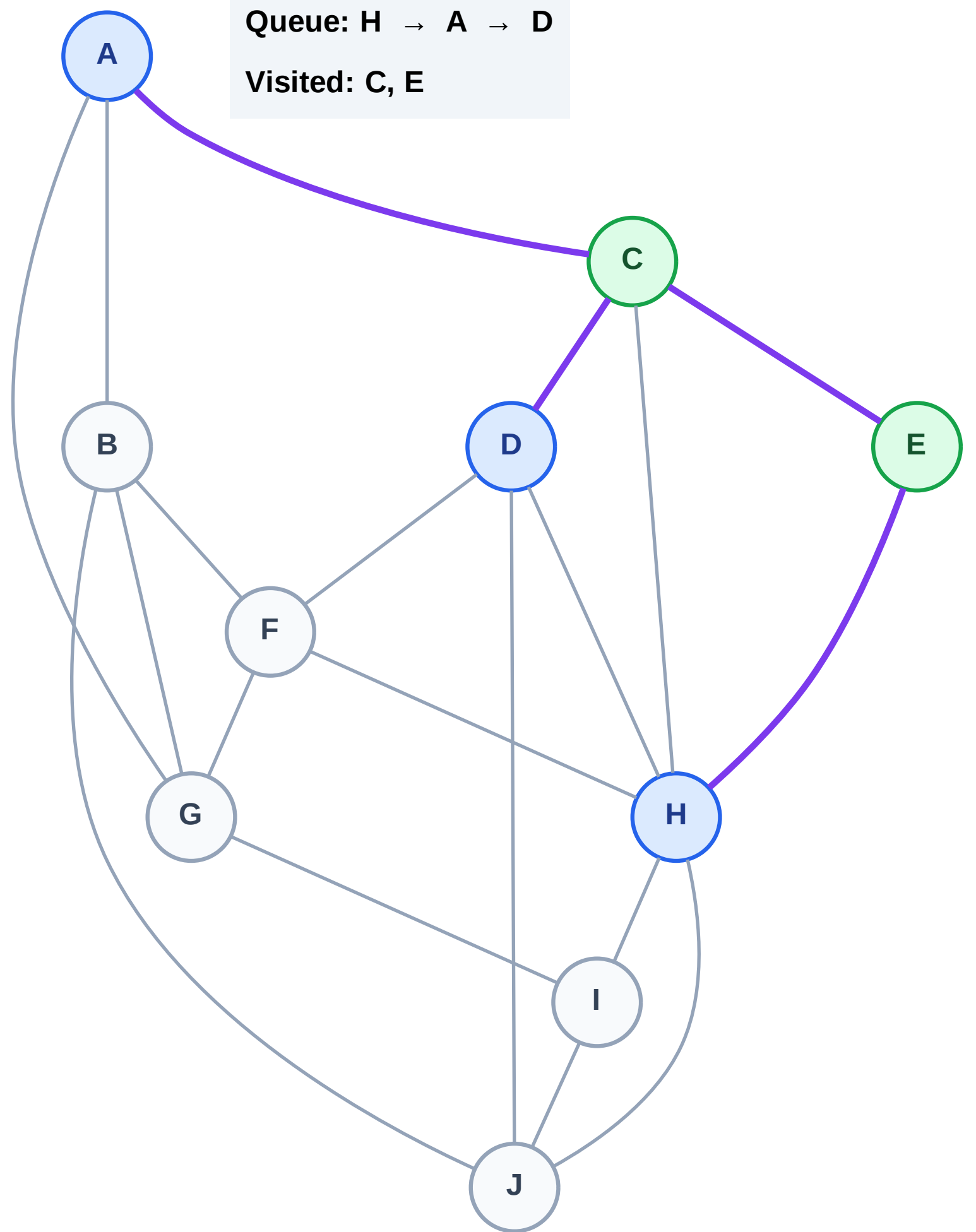
Step 5: Discover A, D from C

Legend



Queue: H → A → D

Visited: C, E



BFS Traversal

Step 6: Process node H

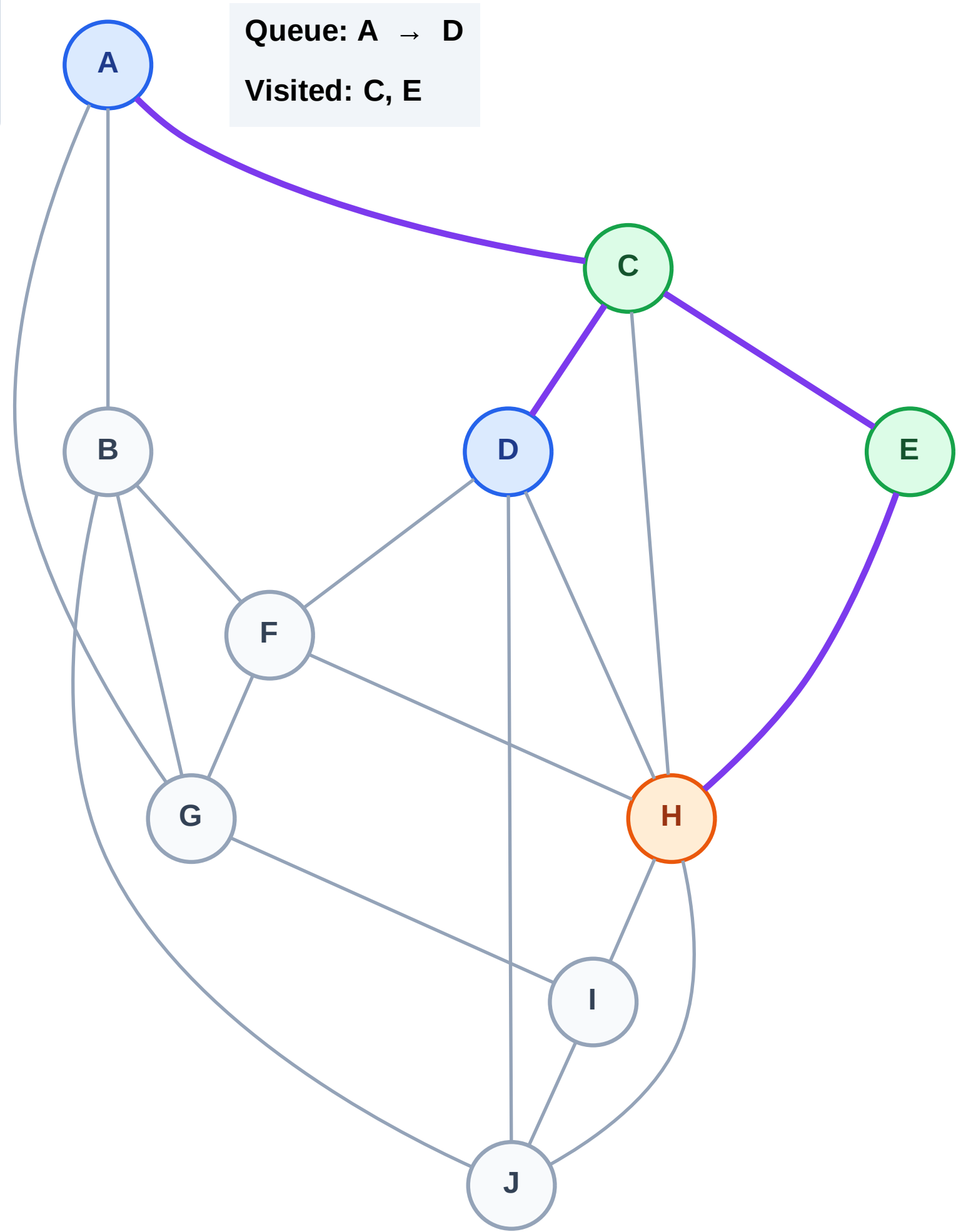
Legend

Complete

Processing

Discovered

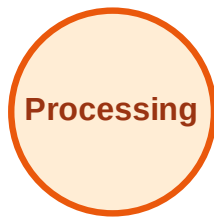
Unseen



BFS Traversal

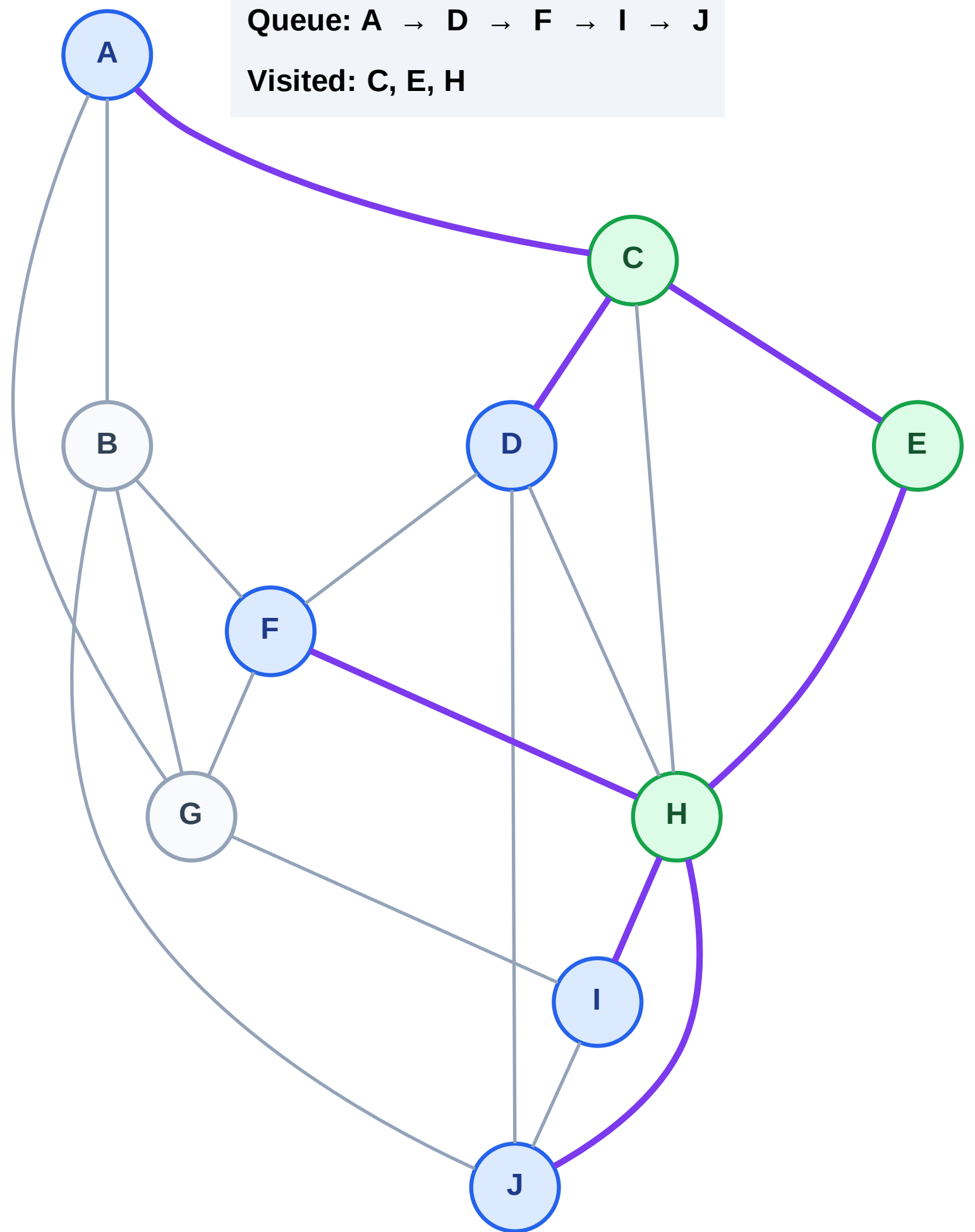
Step 7: Discover F, I, J from H

Legend



Queue: A → D → F → I → J

Visited: C, E, H



BFS Traversal

Step 8: Process node A

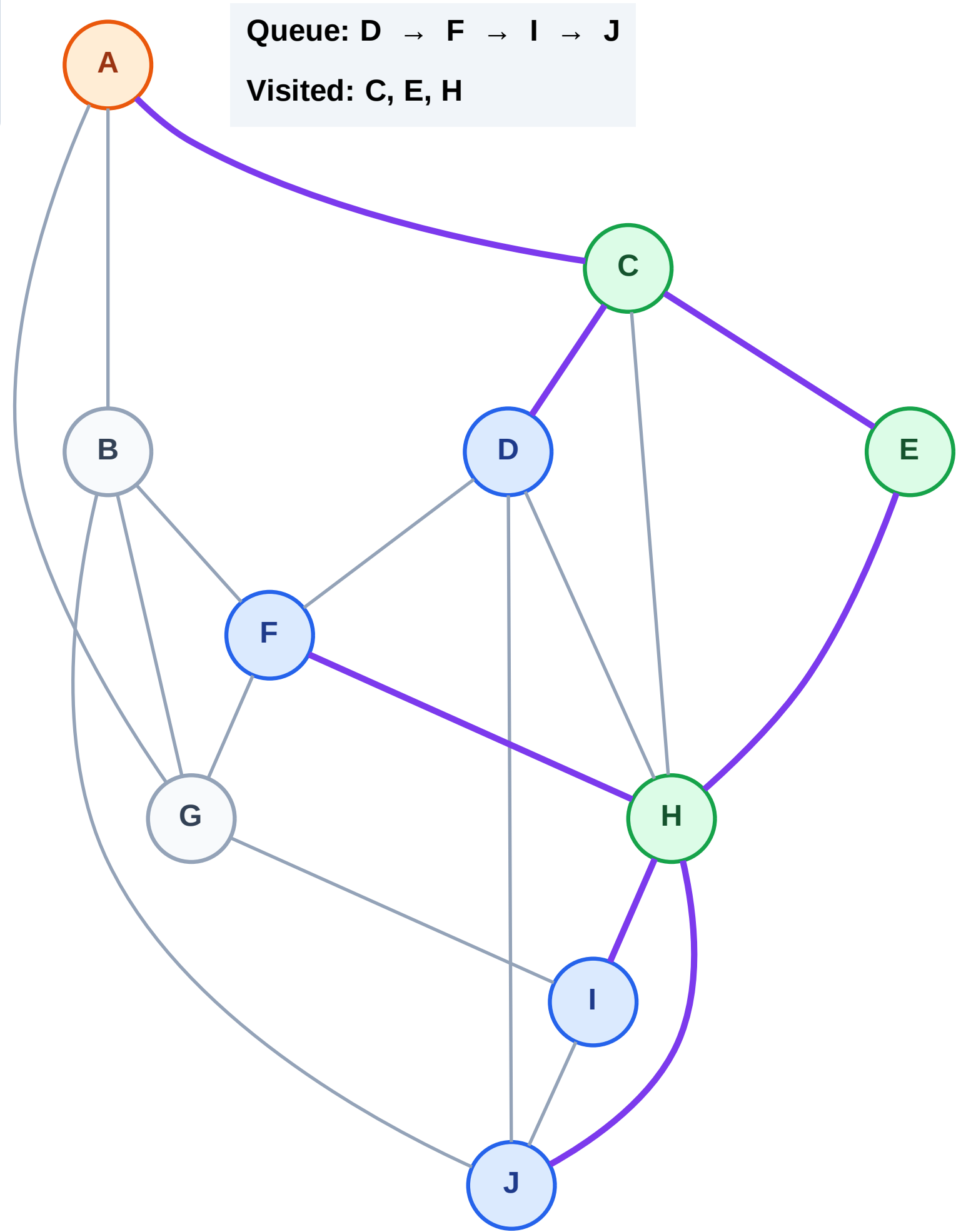
Legend

Complete

Processing

Discovered

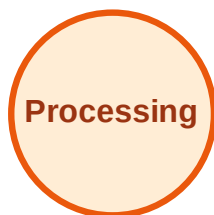
Unseen



BFS Traversal

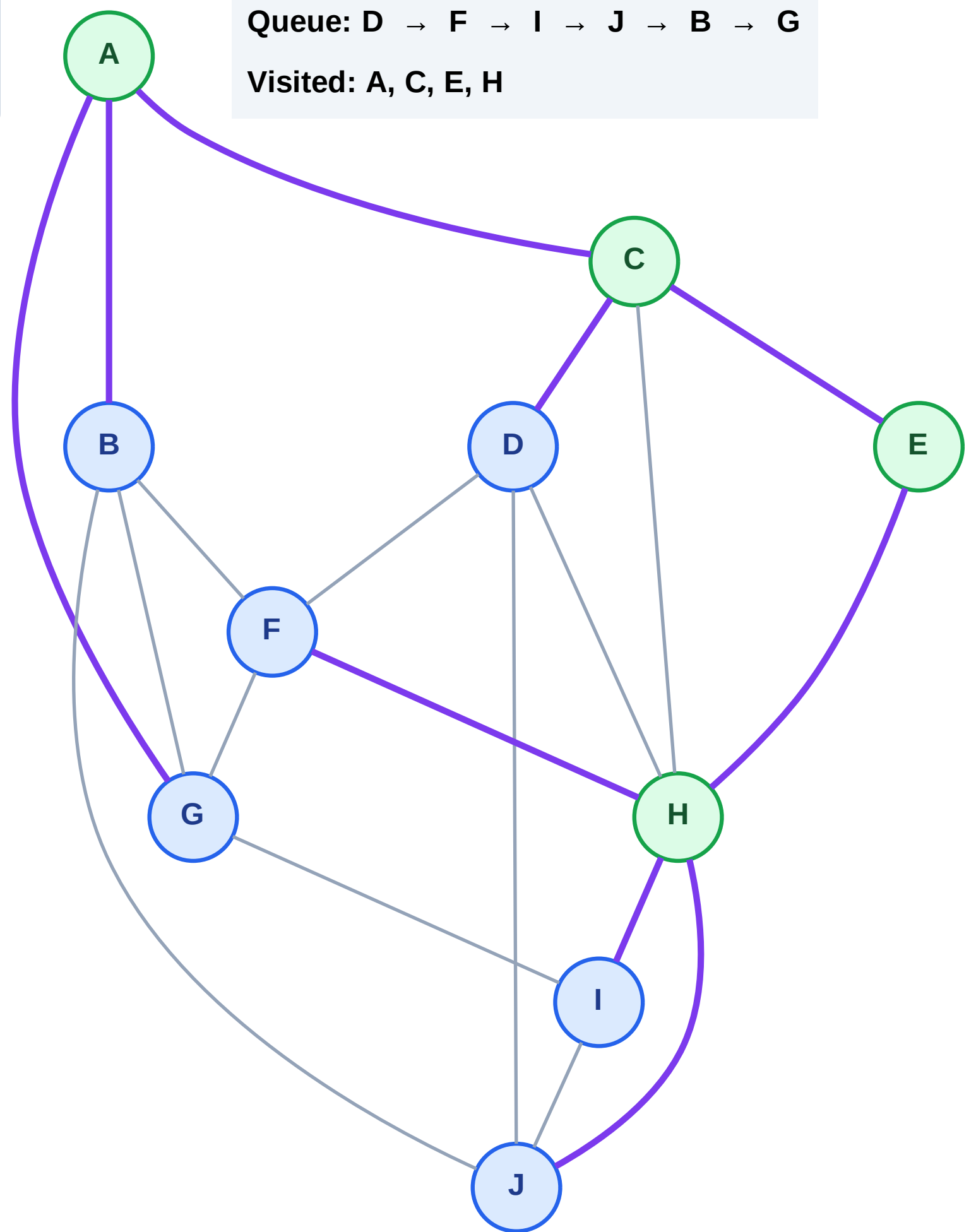
Step 9: Discover B, G from A

Legend



Queue: D → F → I → J → B → G

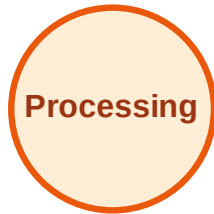
Visited: A, C, E, H



BFS Traversal

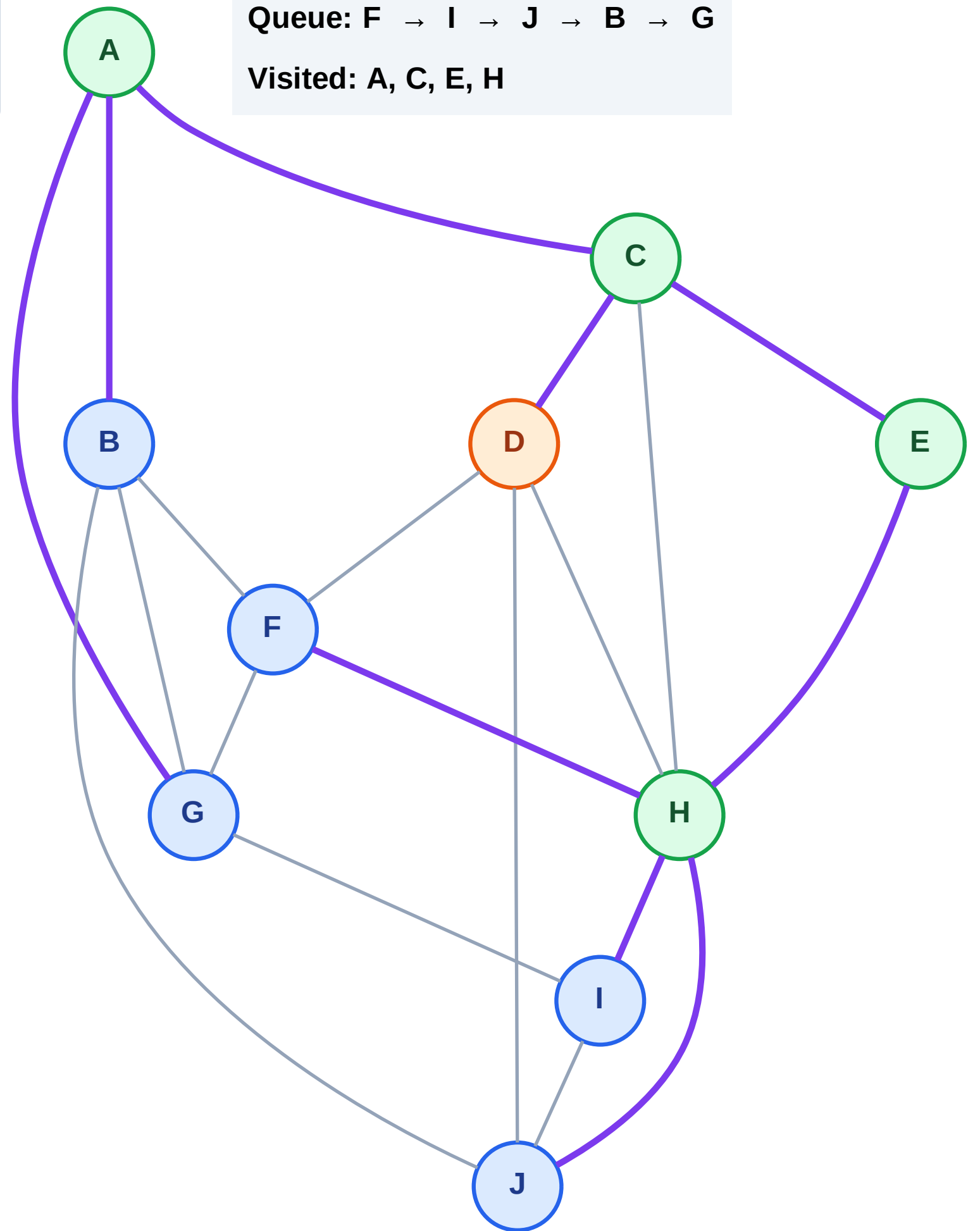
Step 10: Process node D

Legend



Queue: F → I → J → B → G

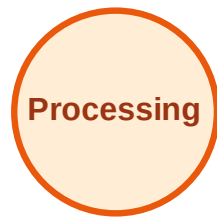
Visited: A, C, E, H



BFS Traversal

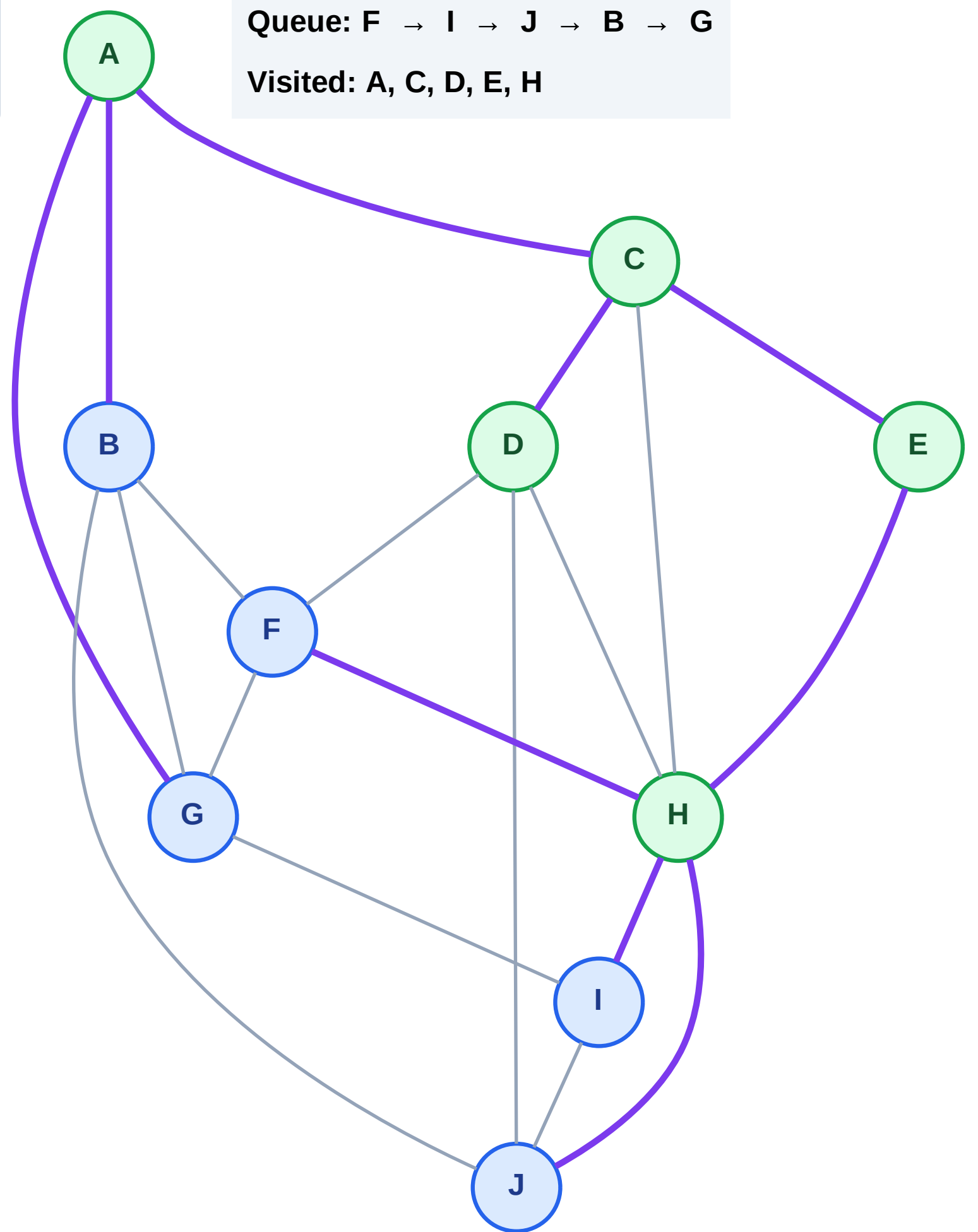
Step 11: No new neighbors from D

Legend



Queue: F → I → J → B → G

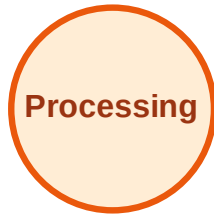
Visited: A, C, D, E, H



BFS Traversal

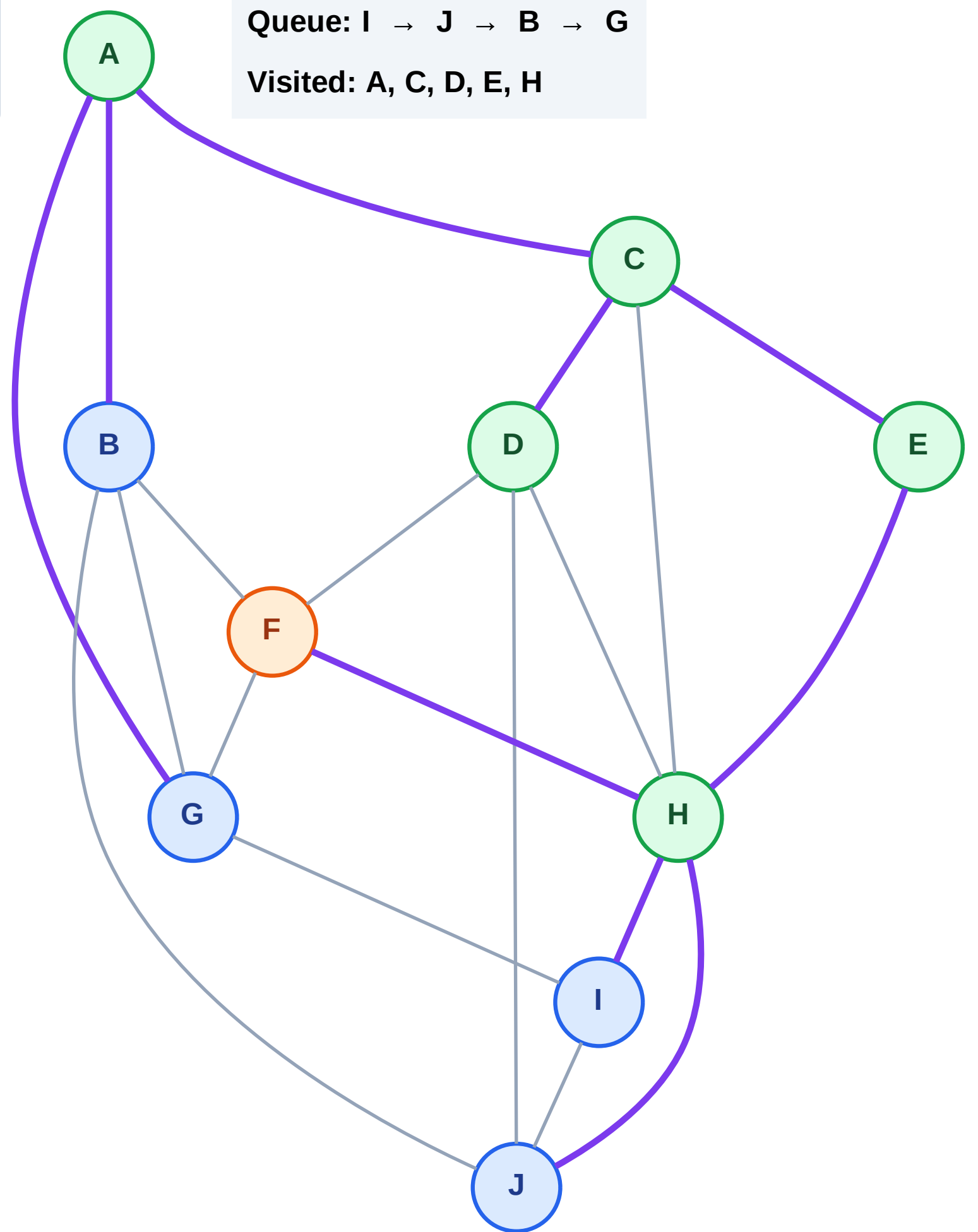
Step 12: Process node F

Legend



Queue: I → J → B → G

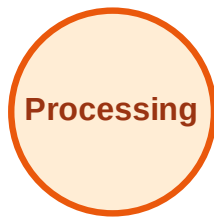
Visited: A, C, D, E, H



BFS Traversal

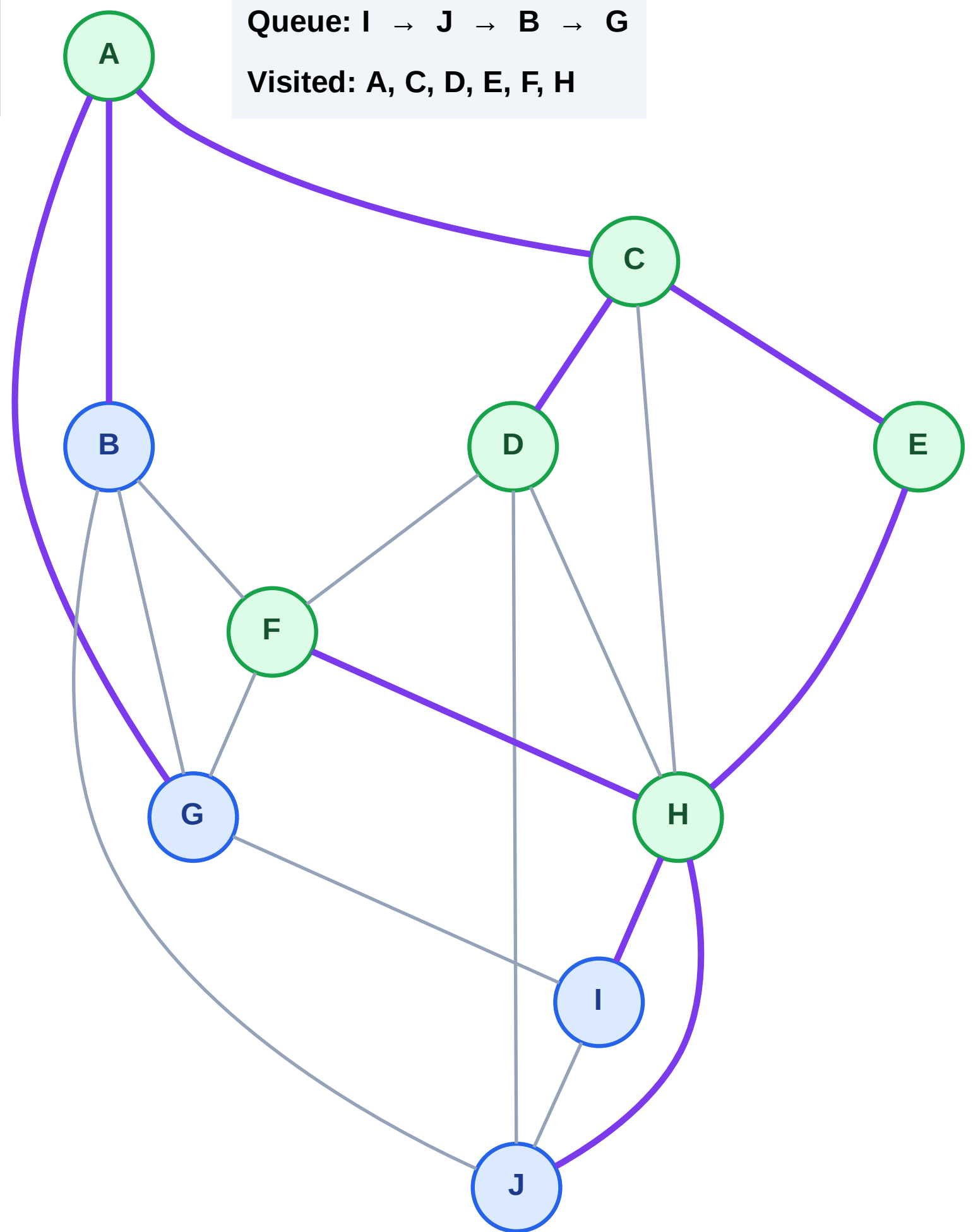
Step 13: No new neighbors from F

Legend



Queue: I → J → B → G

Visited: A, C, D, E, F, H



BFS Traversal

Step 14: Process node I

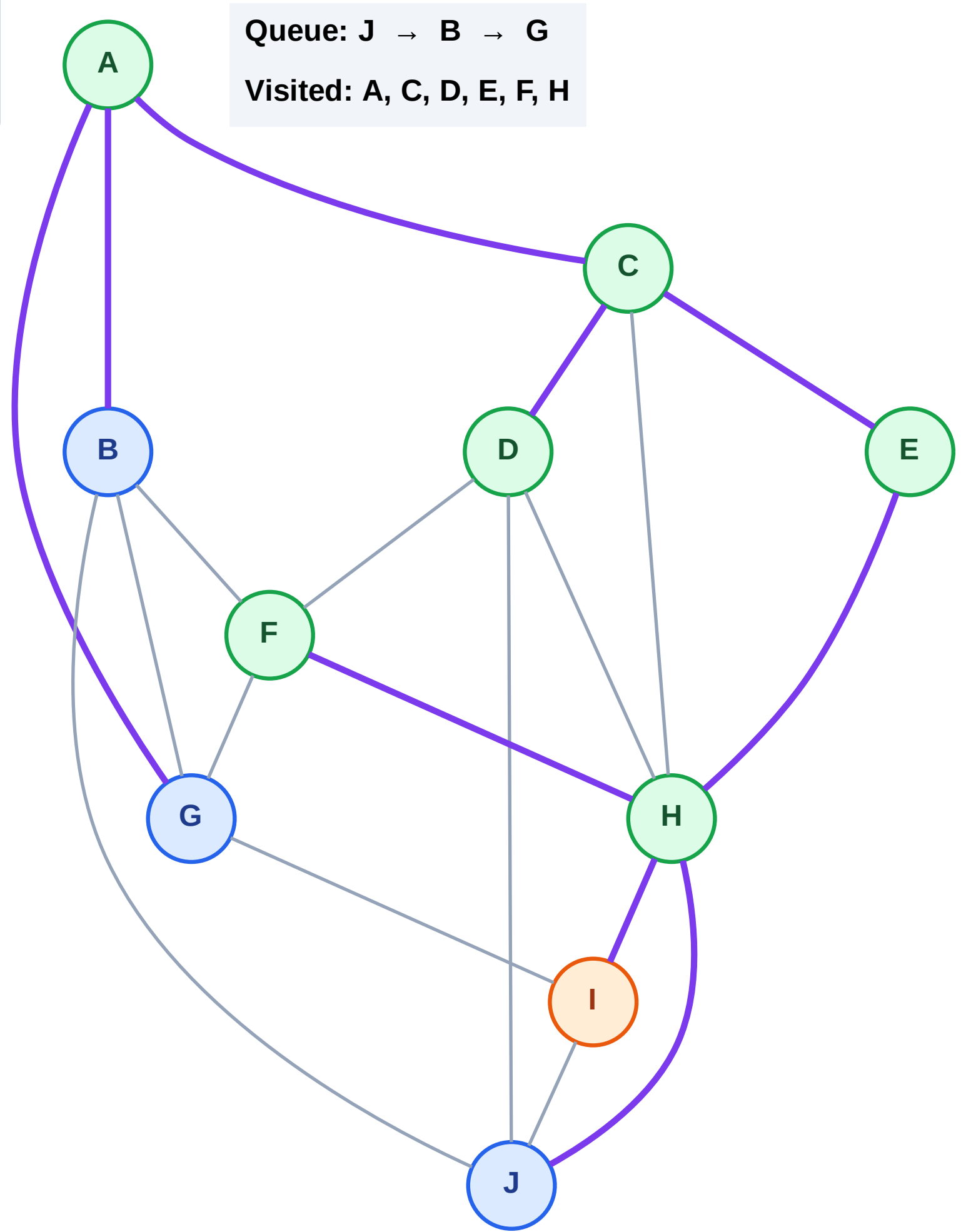
Legend

Complete

Processing

Discovered

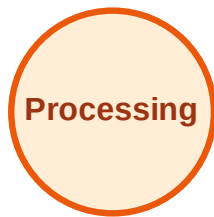
Unseen



BFS Traversal

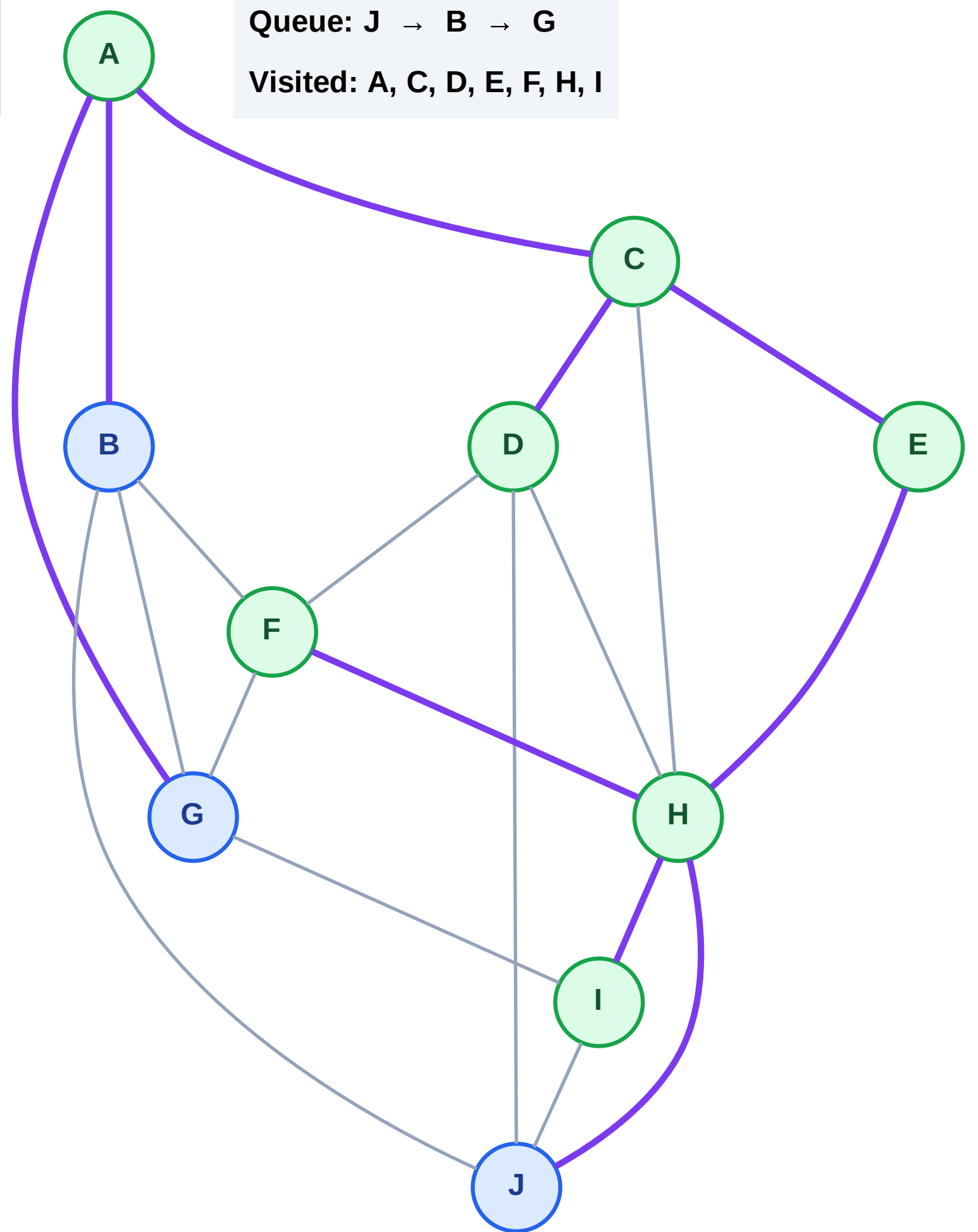
Step 15: No new neighbors from I

Legend



Queue: J → B → G

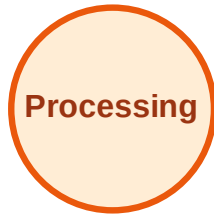
Visited: A, C, D, E, F, H, I



BFS Traversal

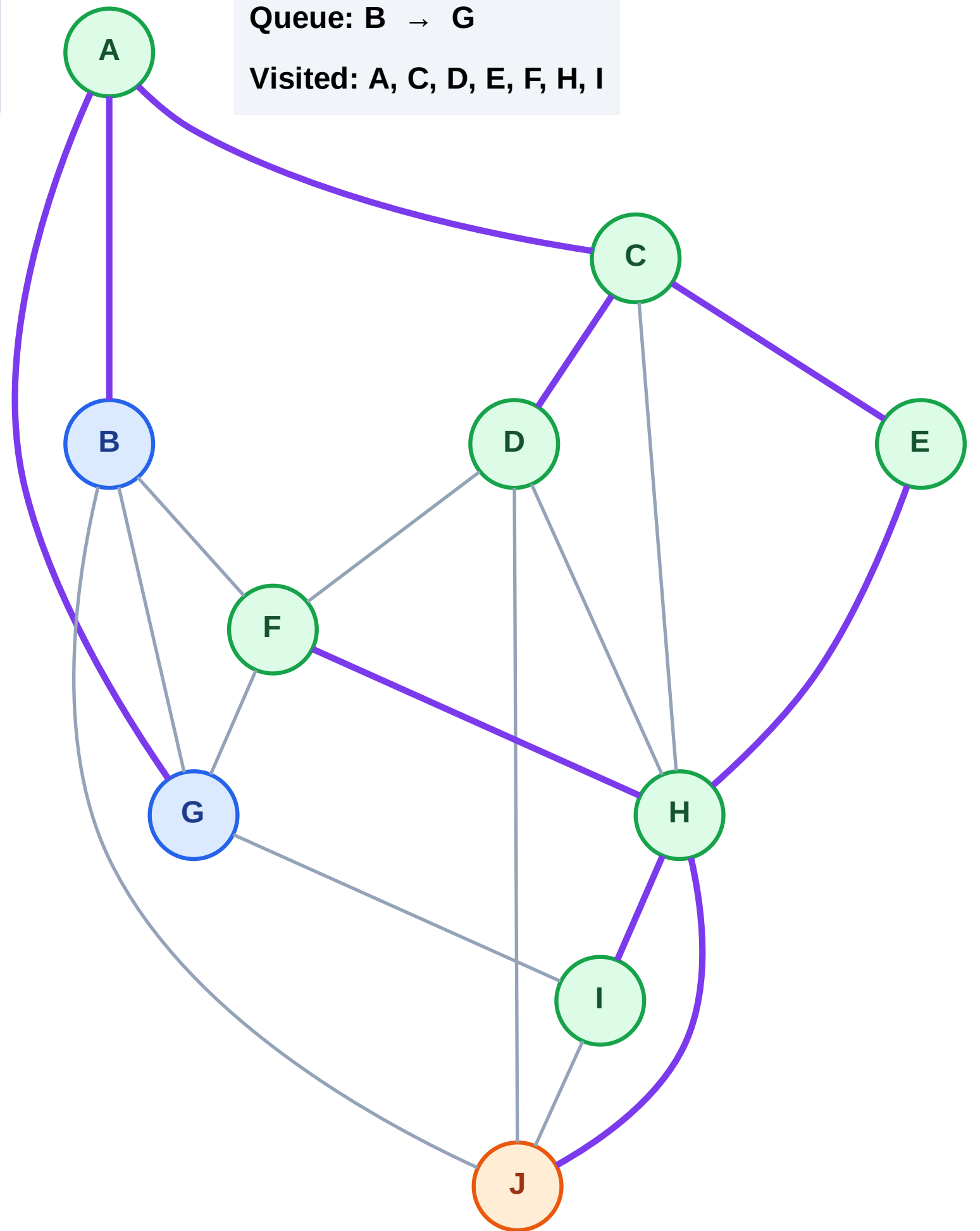
Step 16: Process node J

Legend



Queue: B → G

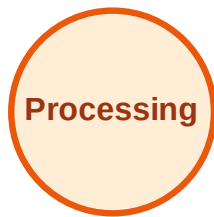
Visited: A, C, D, E, F, H, I



BFS Traversal

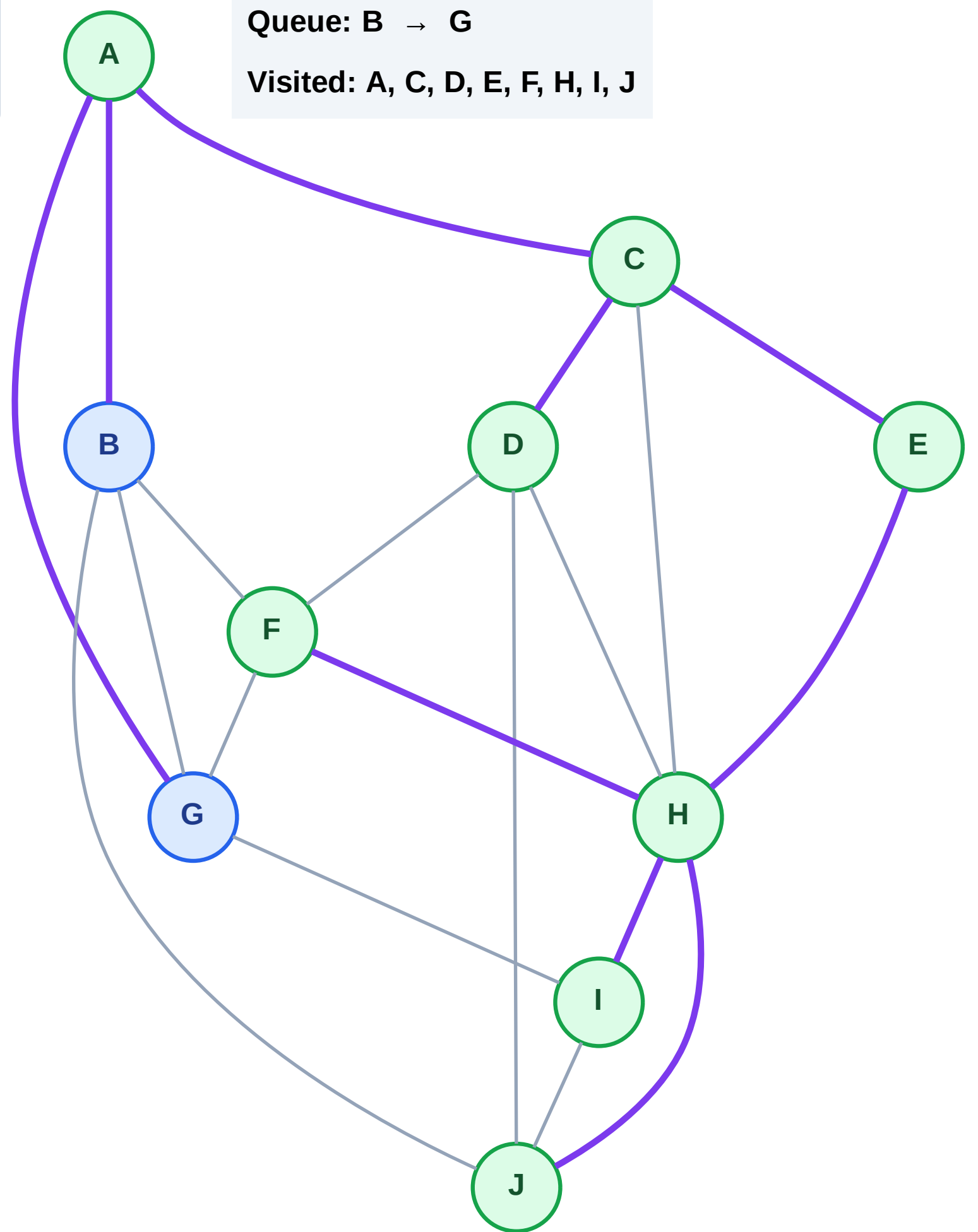
Step 17: No new neighbors from J

Legend



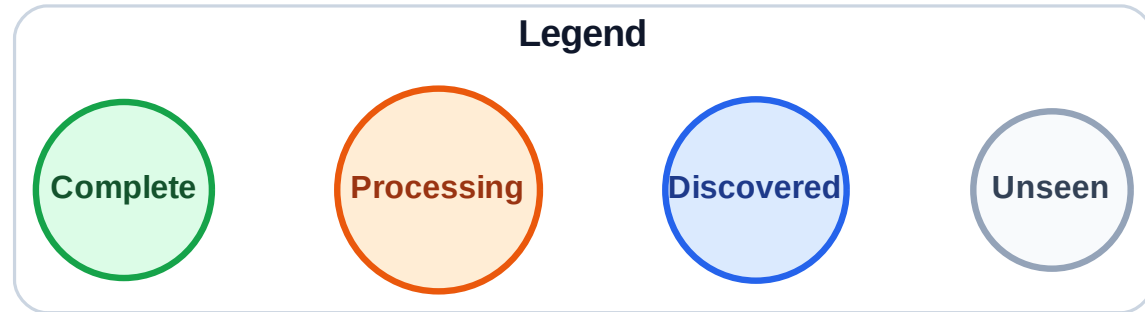
Queue: B → G

Visited: A, C, D, E, F, H, I, J



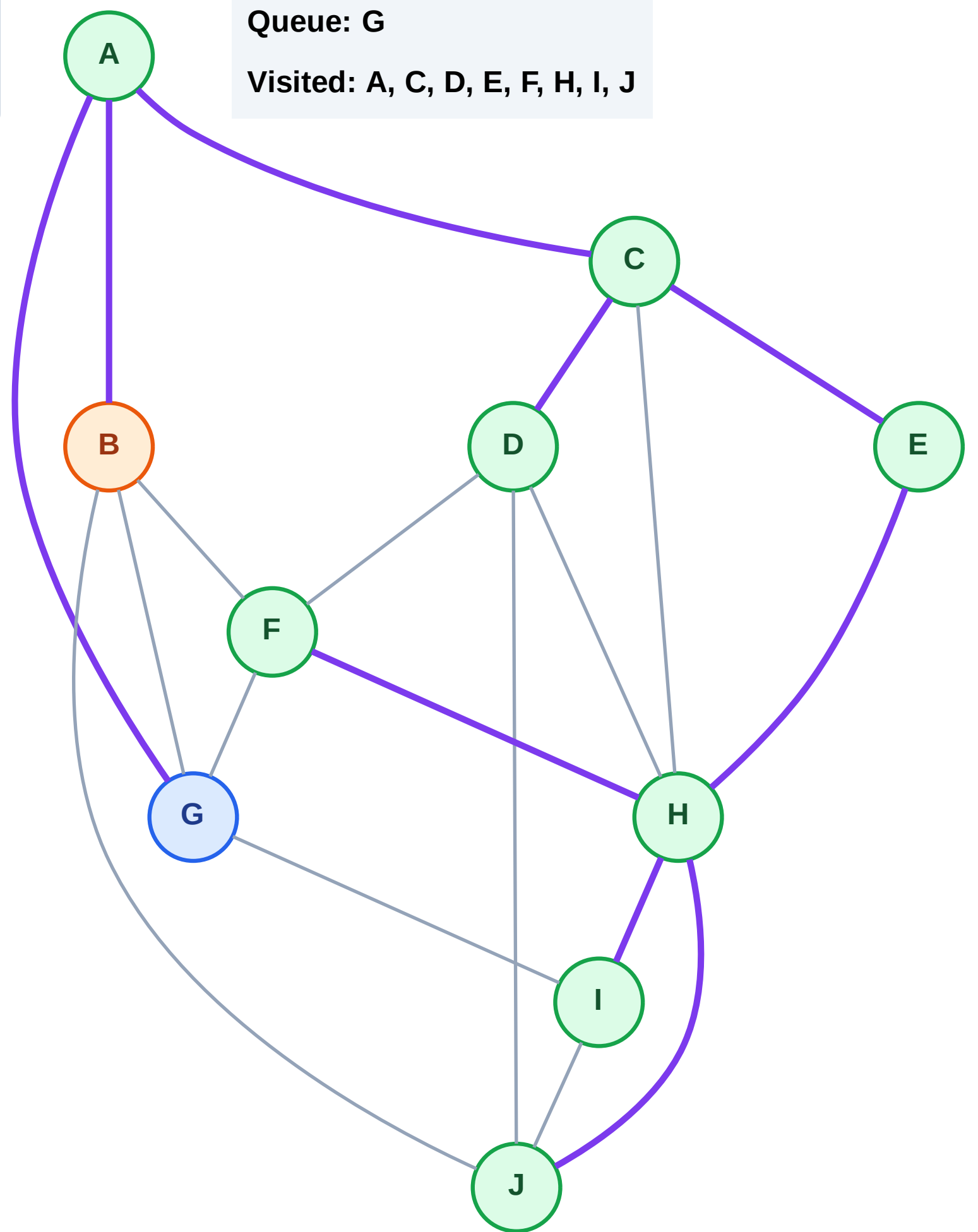
Step 18: Process node B

Step 18: Process node B



Queue: G

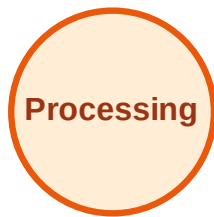
Visited: A, C, D, E, F, H, I, J



BFS Traversal

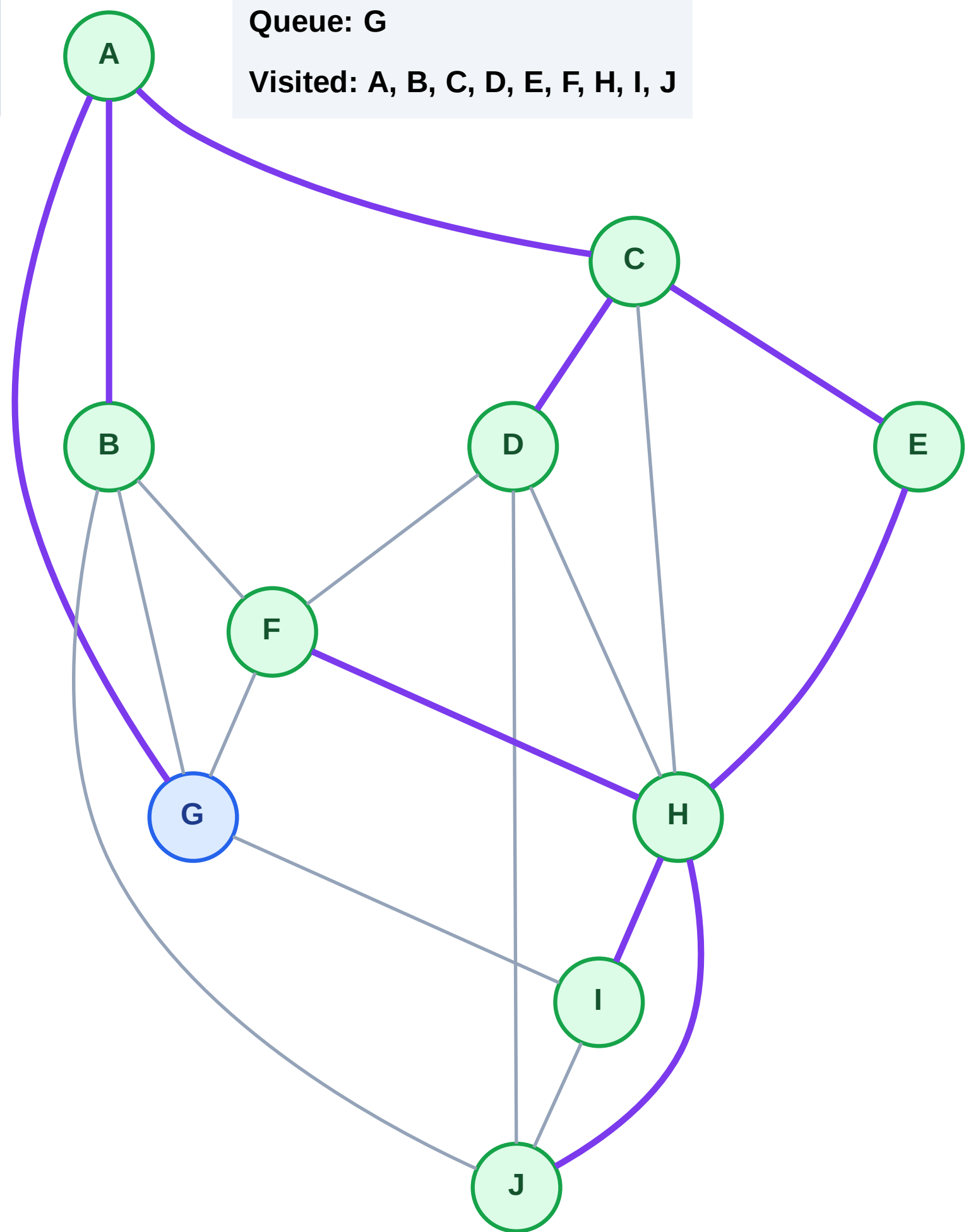
Step 19: No new neighbors from B

Legend



Queue: G

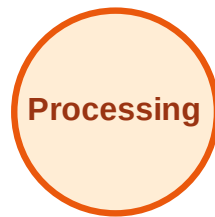
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

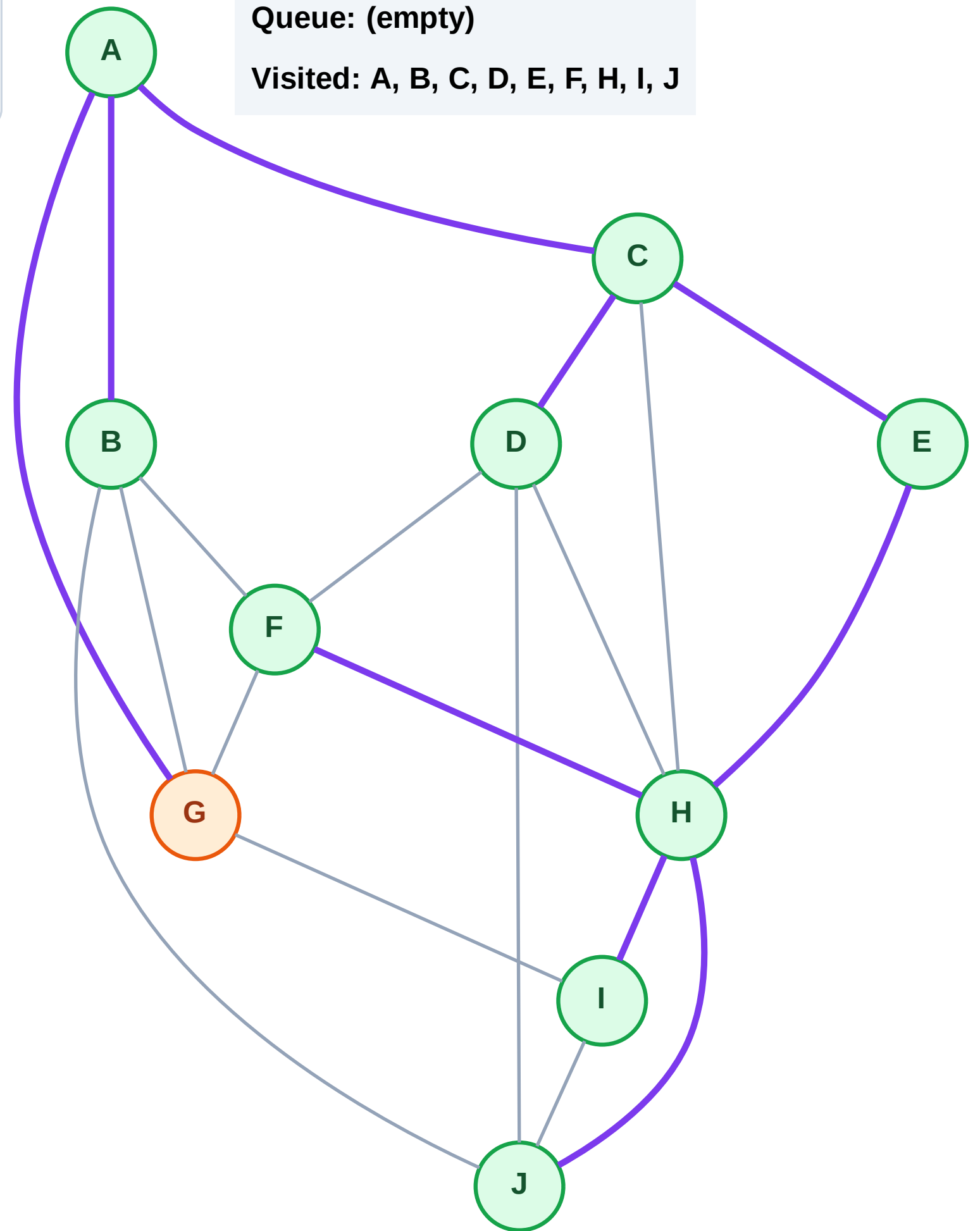
Step 20: Process node G

Legend



Queue: (empty)

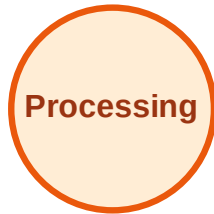
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

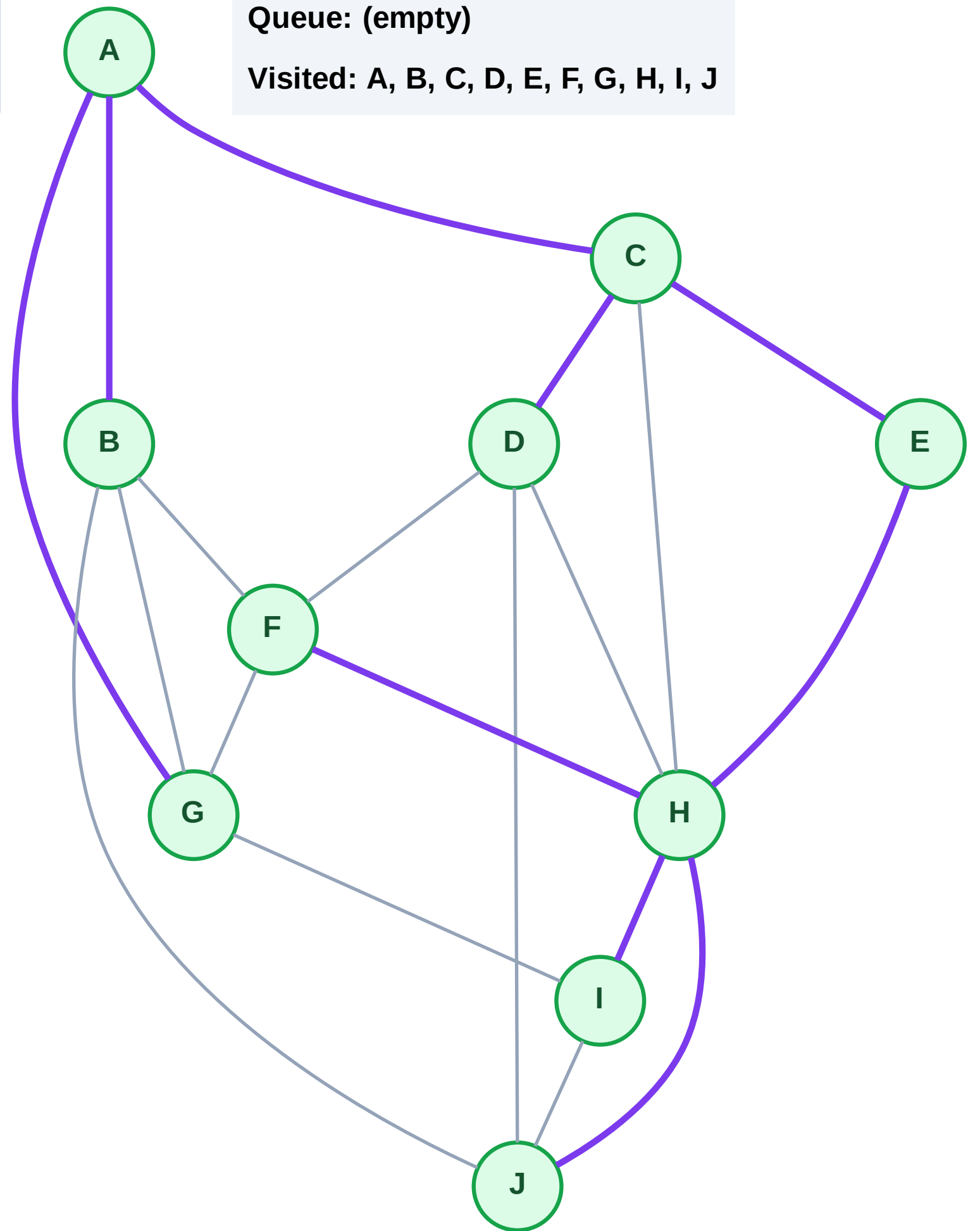
Step 21: No new neighbors from G

Legend



Queue: (empty)

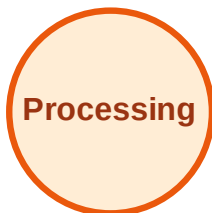
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

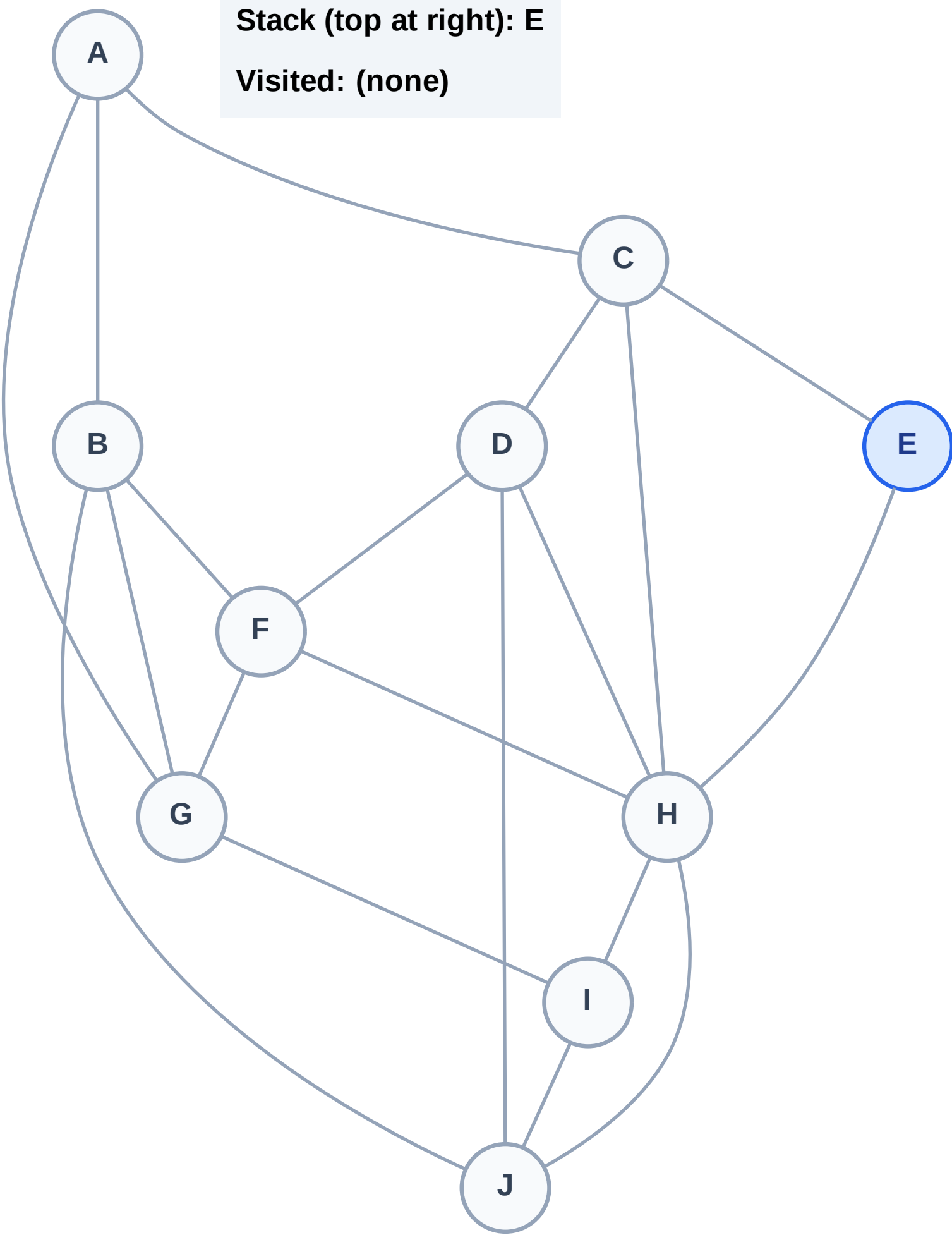
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

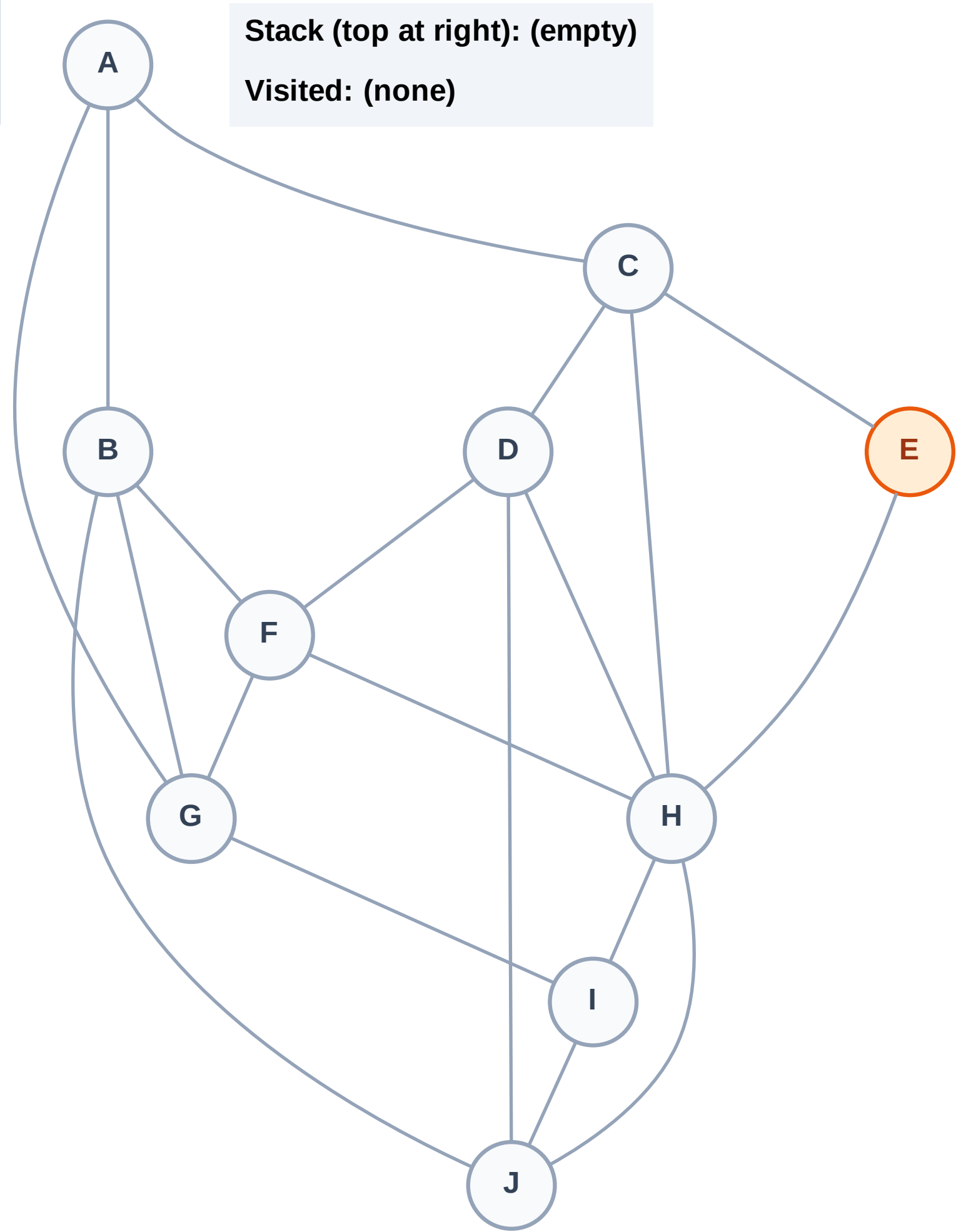
Legend

Complete

Processing

Discovered

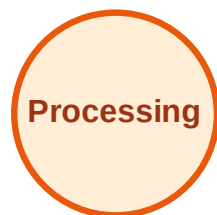
Unseen



DFS Traversal

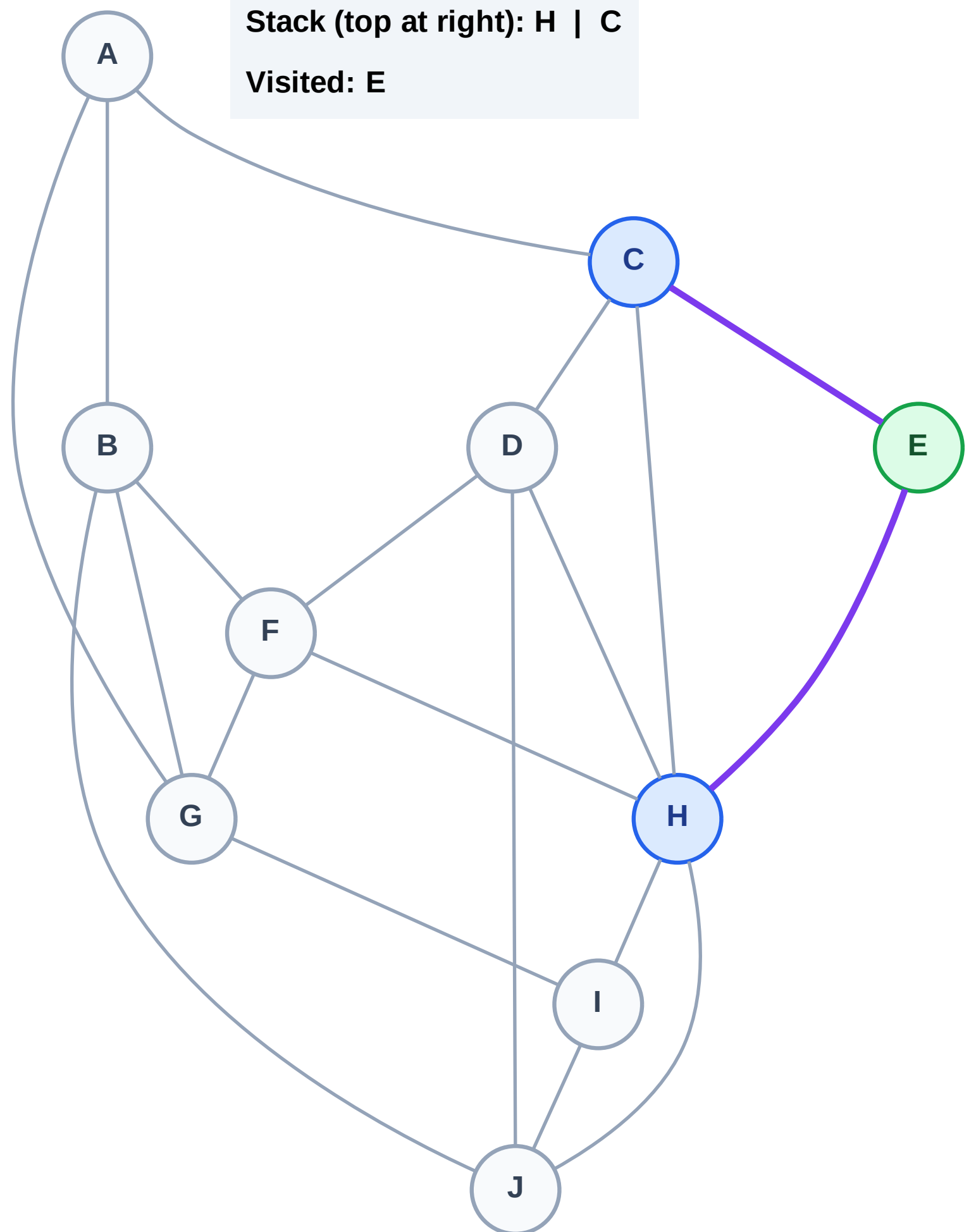
Step 3: Discover H, C from E

Legend



Stack (top at right): H | C

Visited: E



DFS Traversal

Step 4: Process node C

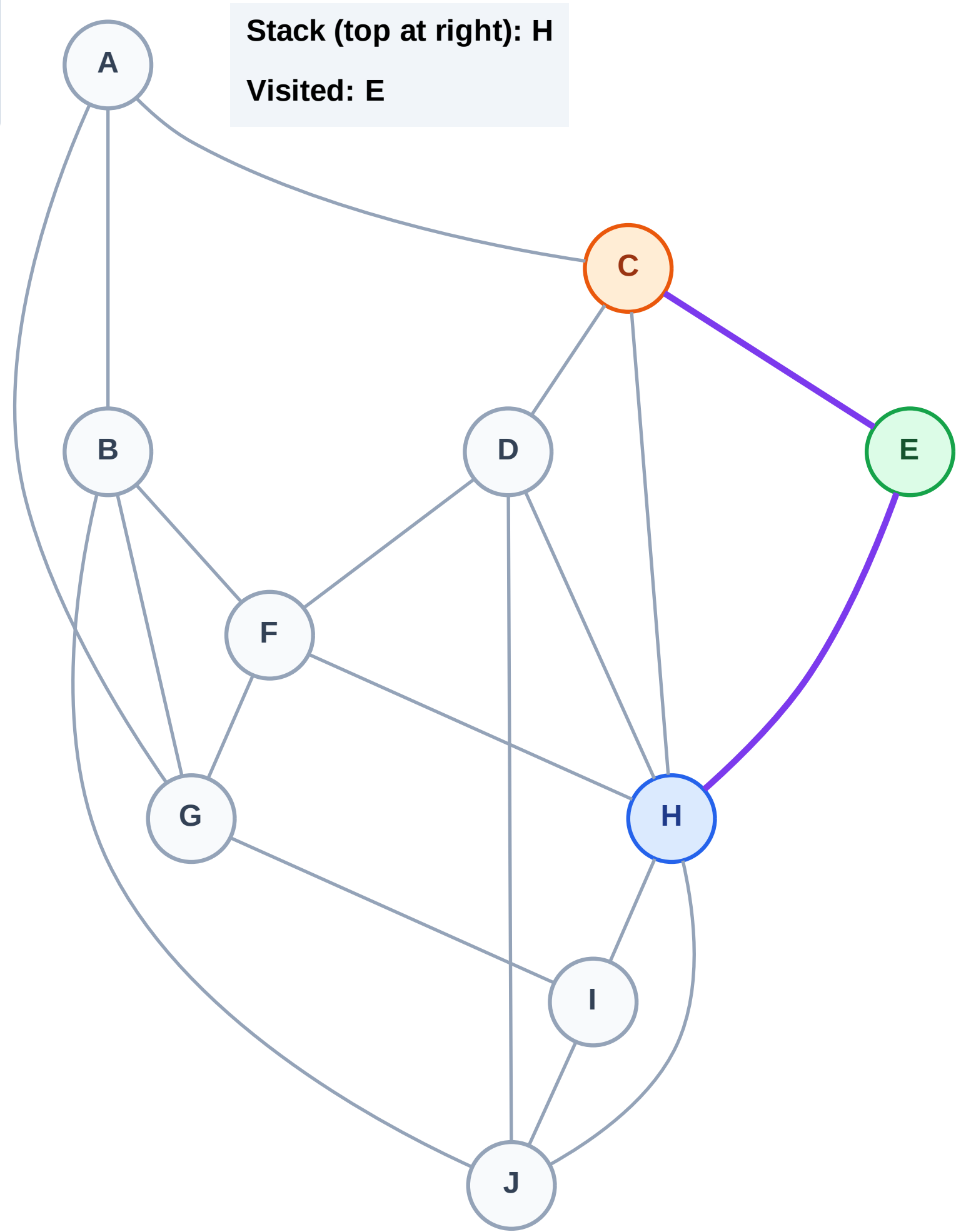
Legend

Complete

Processing

Discovered

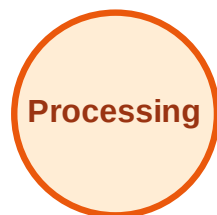
Unseen



DFS Traversal

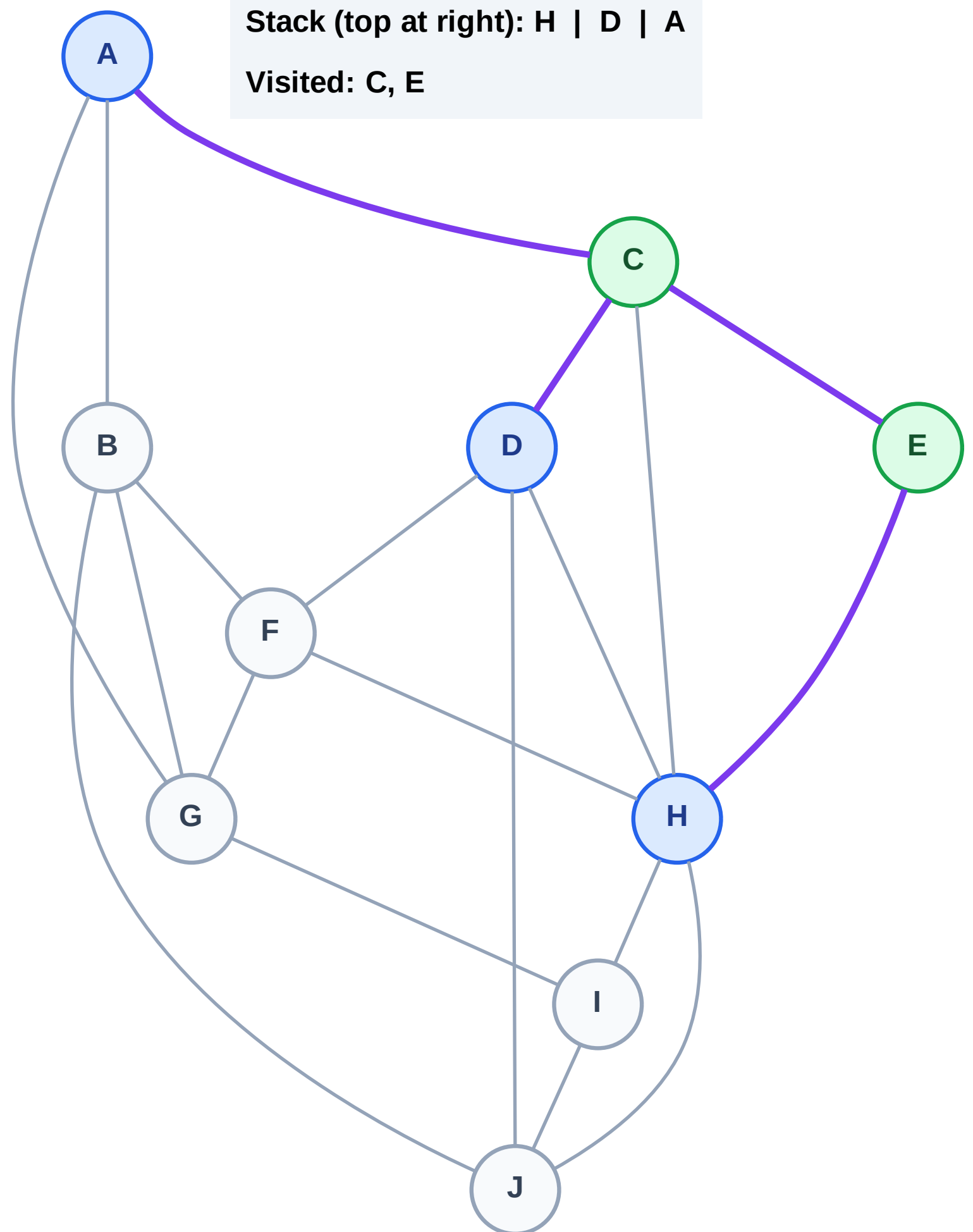
Step 5: Discover D, A from C

Legend



Stack (top at right): H | D | A

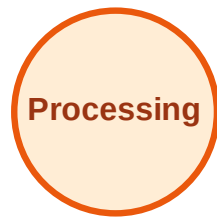
Visited: C, E



DFS Traversal

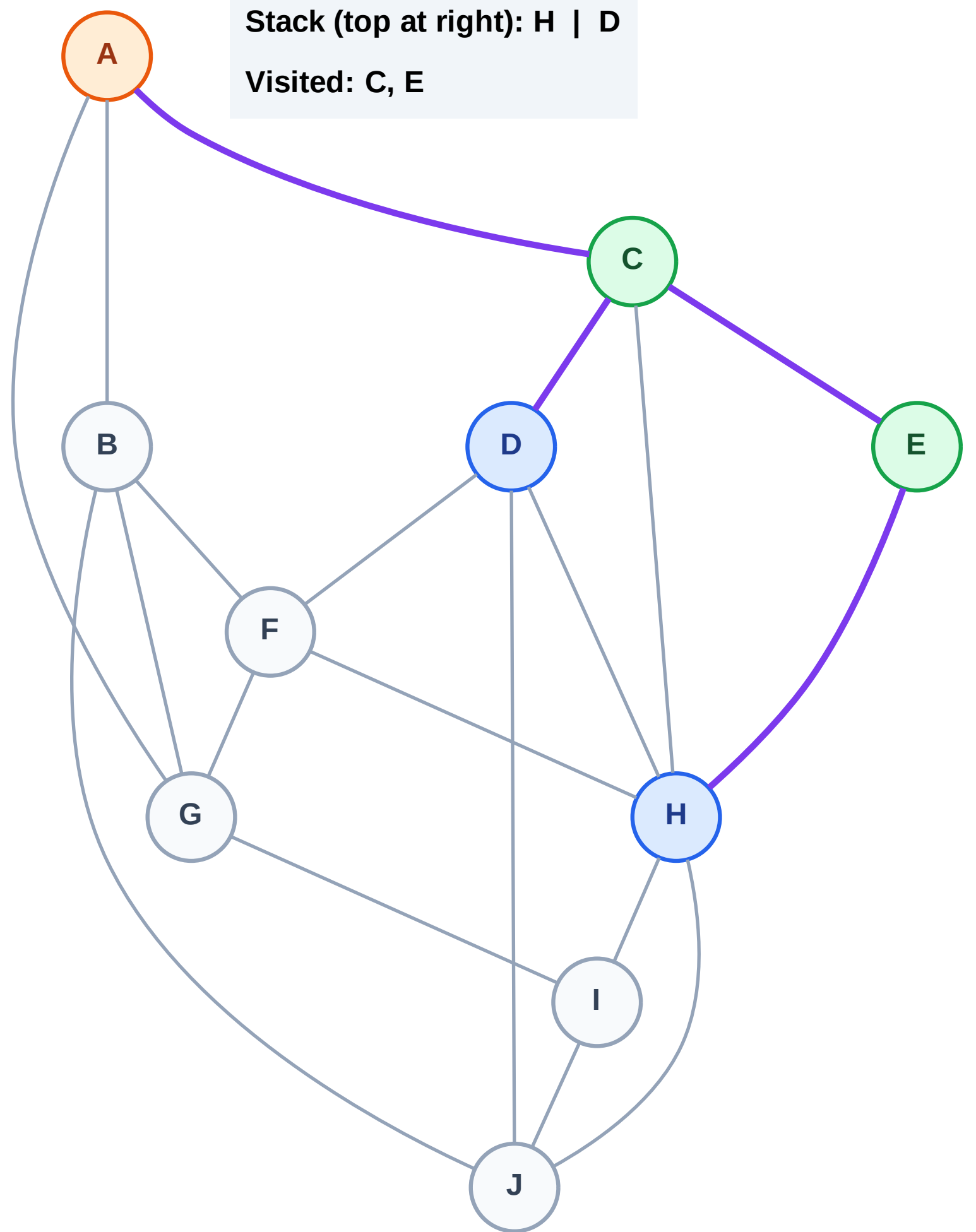
Step 6: Process node A

Legend



Stack (top at right): H | D

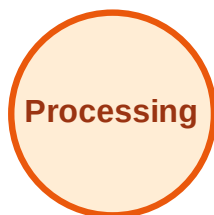
Visited: C, E



DFS Traversal

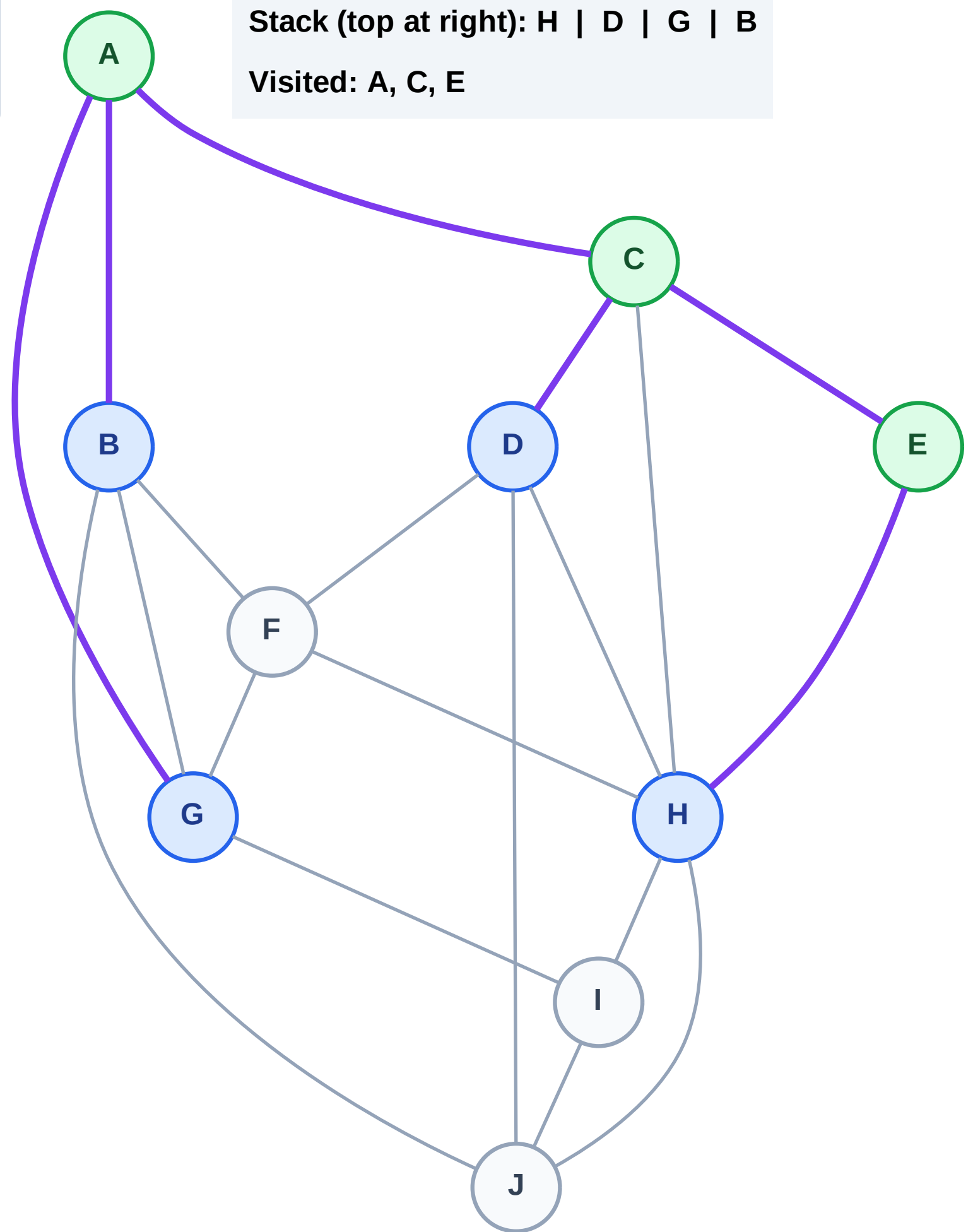
Step 7: Discover G, B from A

Legend



Stack (top at right): H | D | G | B

Visited: A, C, E



DFS Traversal

Step 8: Process node B

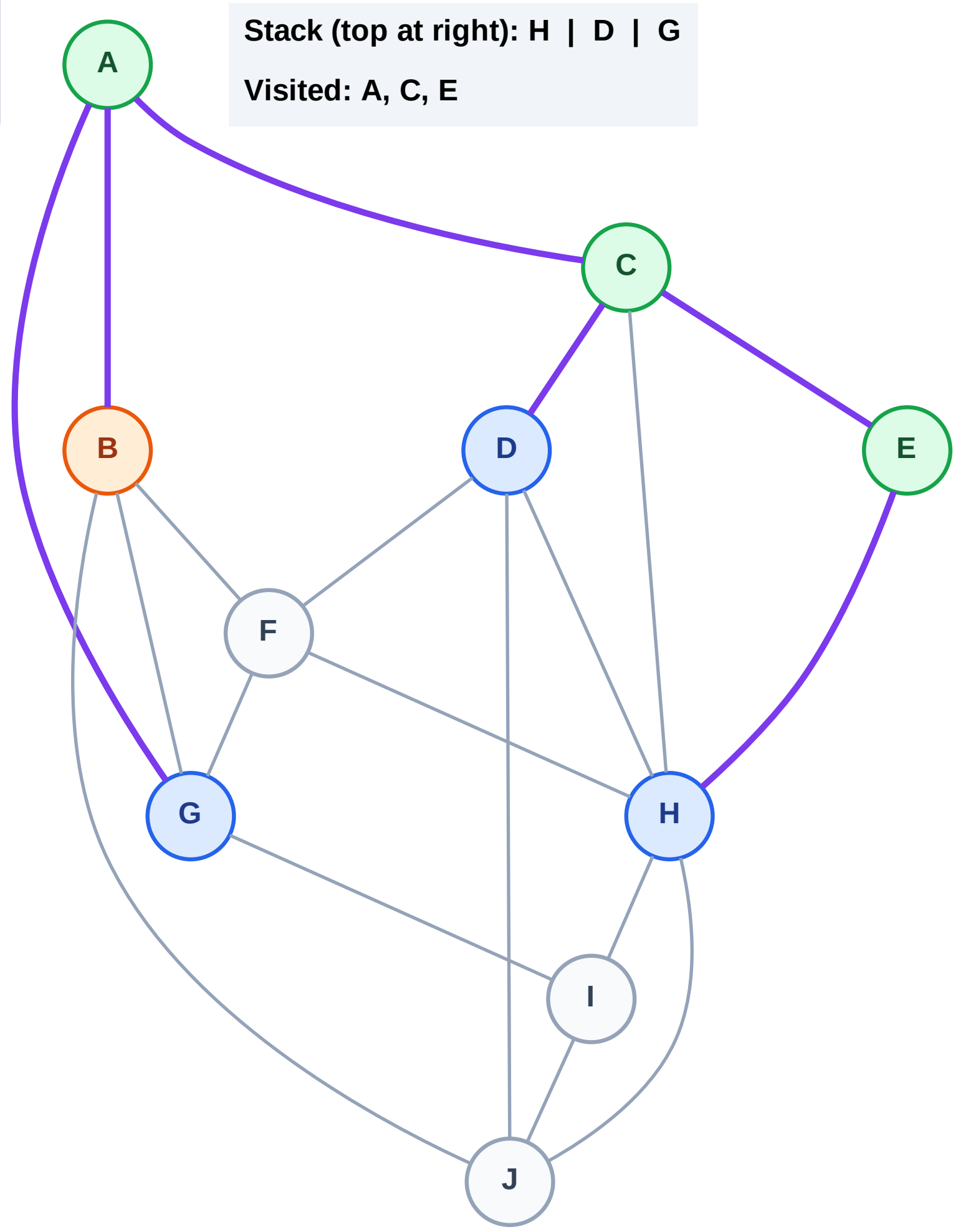
Legend

Complete

Processing

Discovered

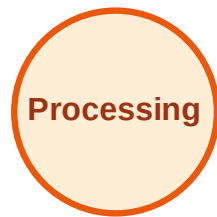
Unseen



DFS Traversal

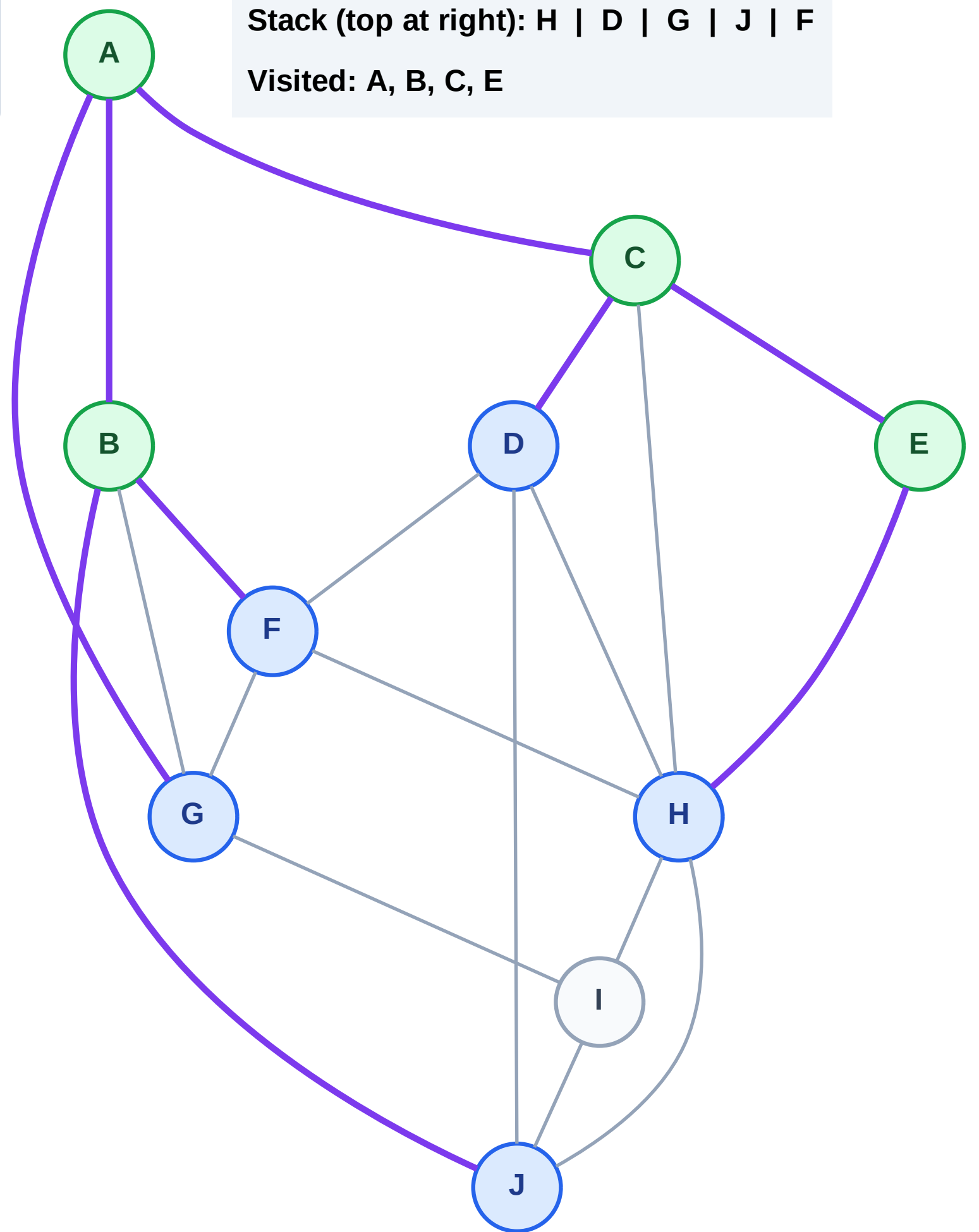
Step 9: Discover J, F from B

Legend



Stack (top at right): H | D | G | J | F

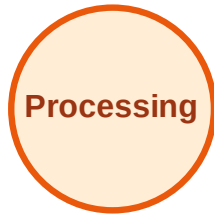
Visited: A, B, C, E



DFS Traversal

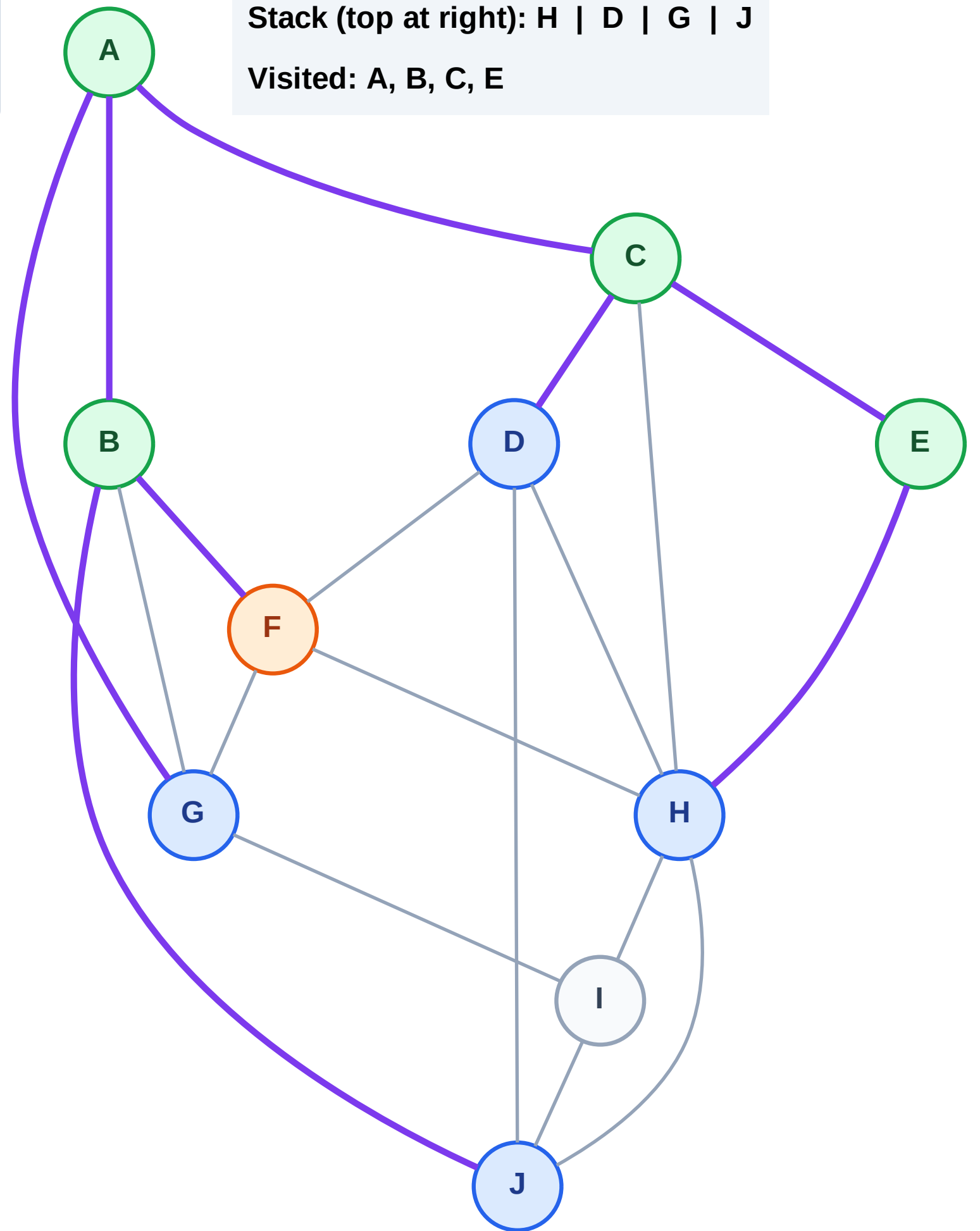
Step 10: Process node F

Legend



Stack (top at right): H | D | G | J

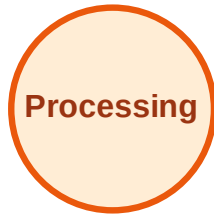
Visited: A, B, C, E



DFS Traversal

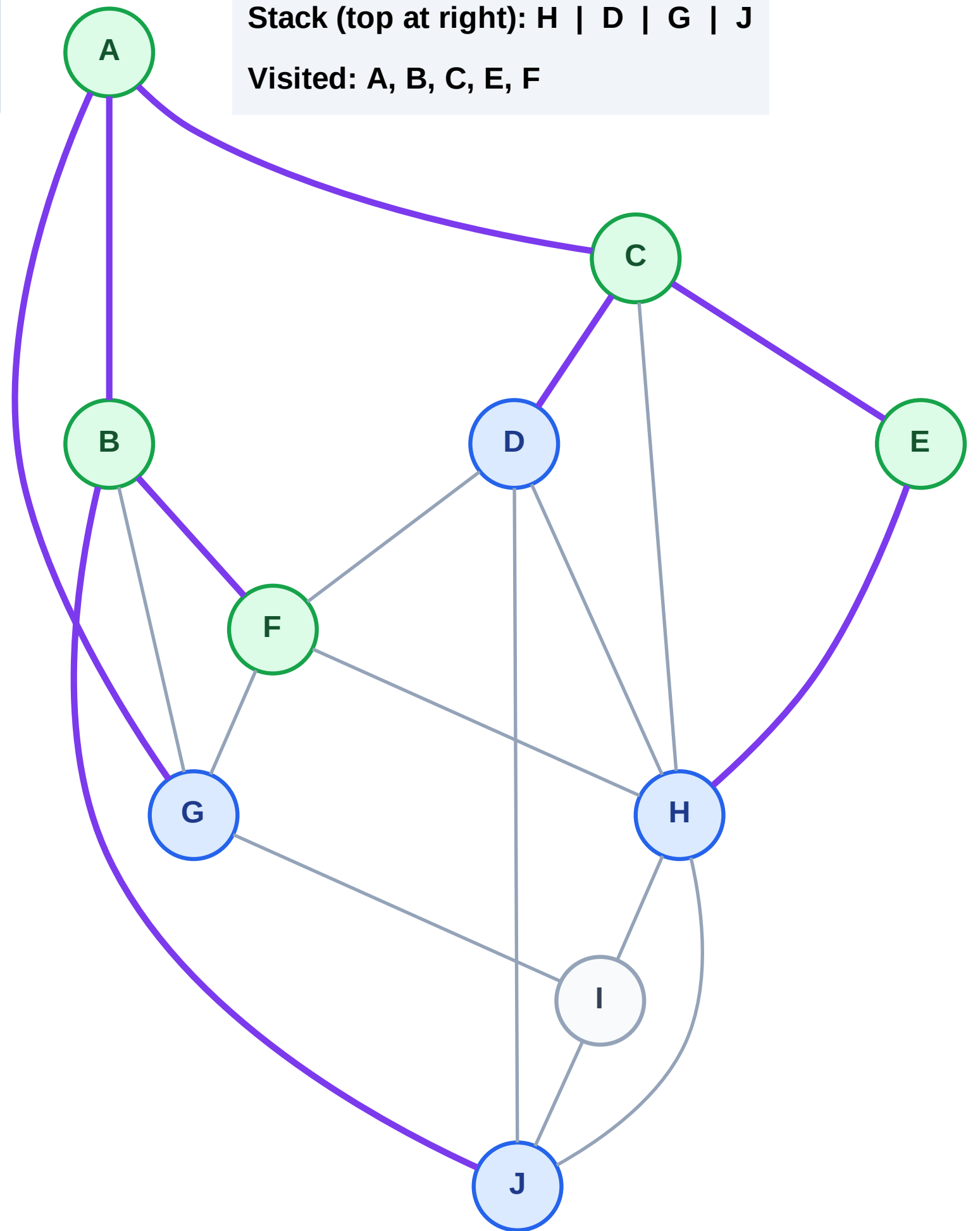
Step 11: No new neighbors from F

Legend



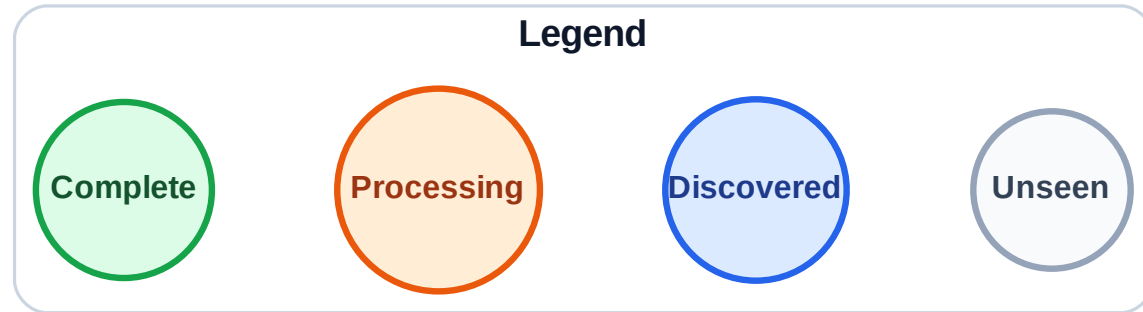
Stack (top at right): H | D | G | J

Visited: A, B, C, E, F



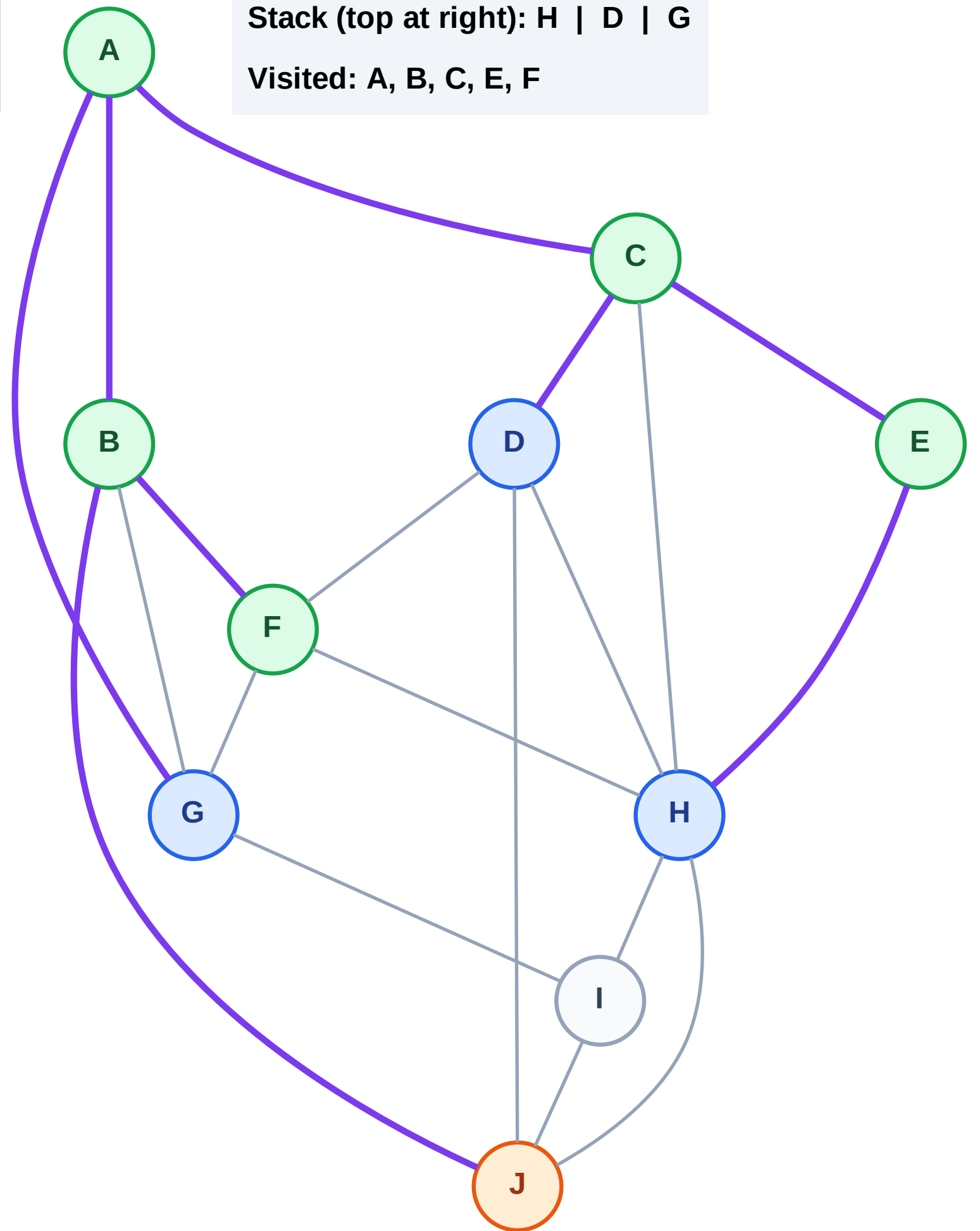
Step 12: Process node J

Step 12: Process node J



Stack (top at right): H | D | G

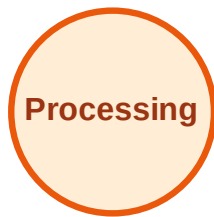
Visited: A, B, C, E, F



DFS Traversal

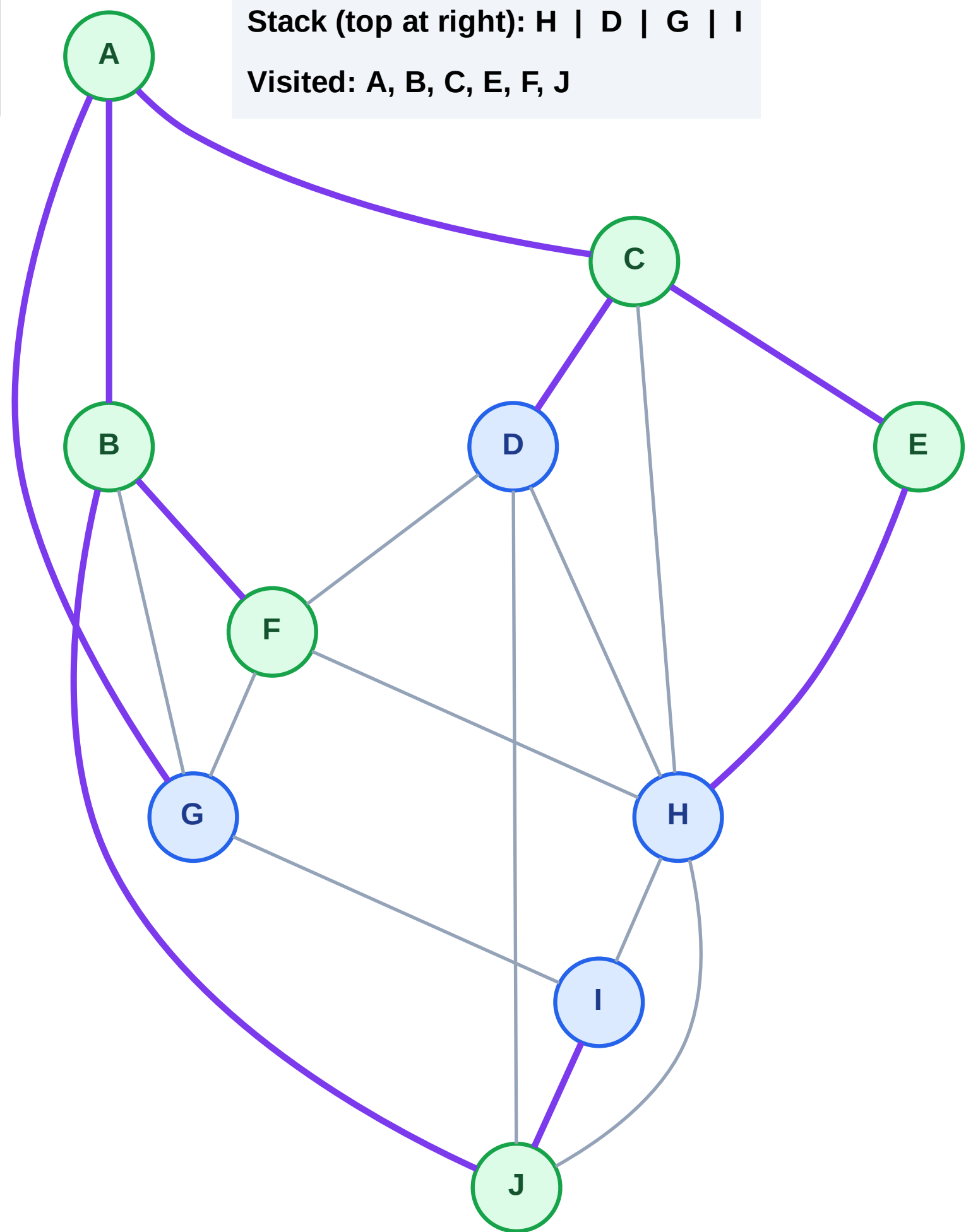
Step 13: Discover I from J

Legend



Stack (top at right): H | D | G | I

Visited: A, B, C, E, F, J



DFS Traversal

Step 14: Process node I

Legend

Complete

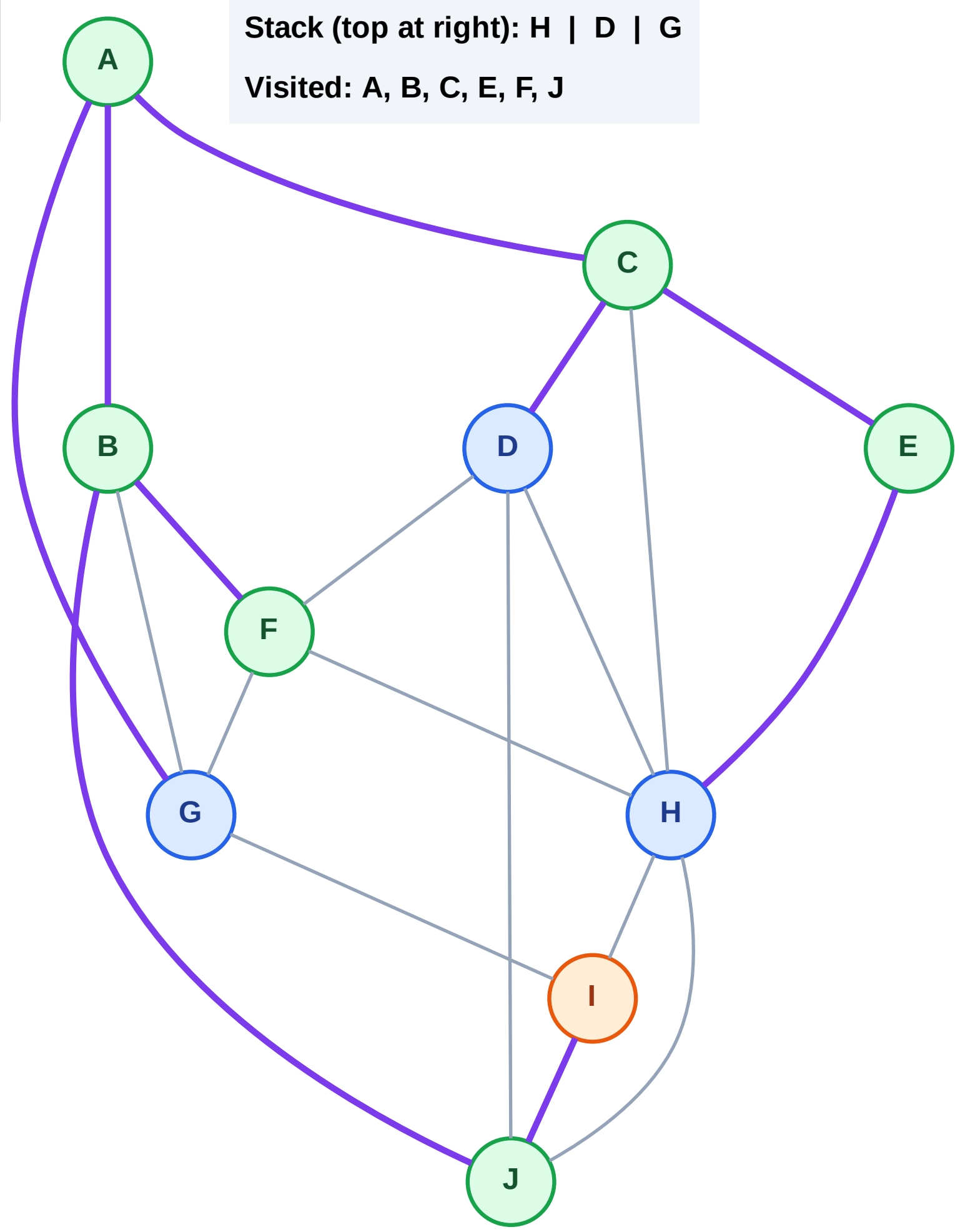
Processing

Discovered

Unseen

Stack (top at right): H | D | G

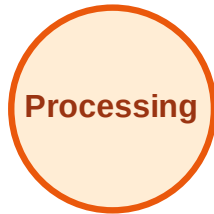
Visited: A, B, C, E, F, J



DFS Traversal

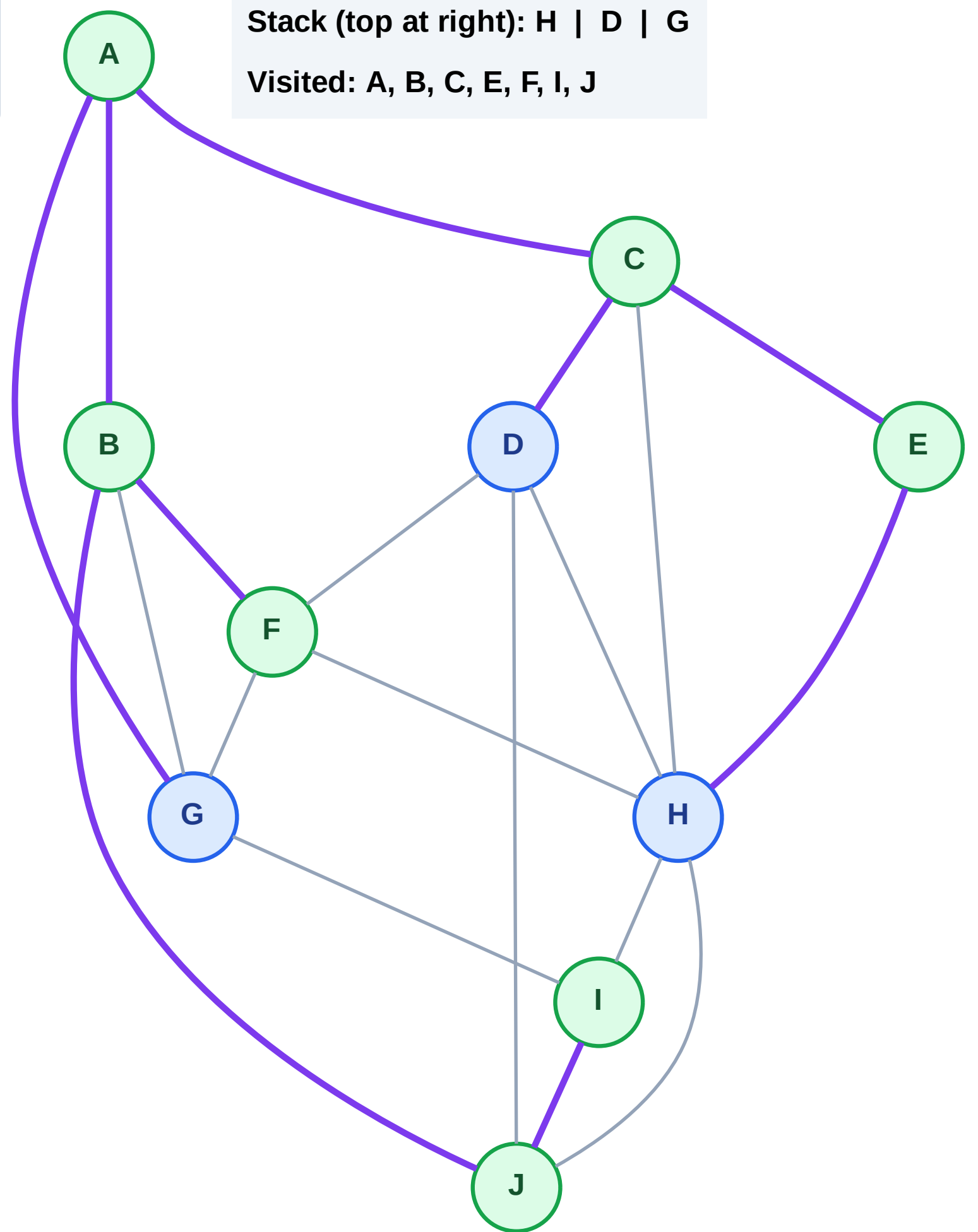
Step 15: No new neighbors from I

Legend



Stack (top at right): H | D | G

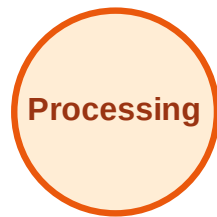
Visited: A, B, C, E, F, I, J



DFS Traversal

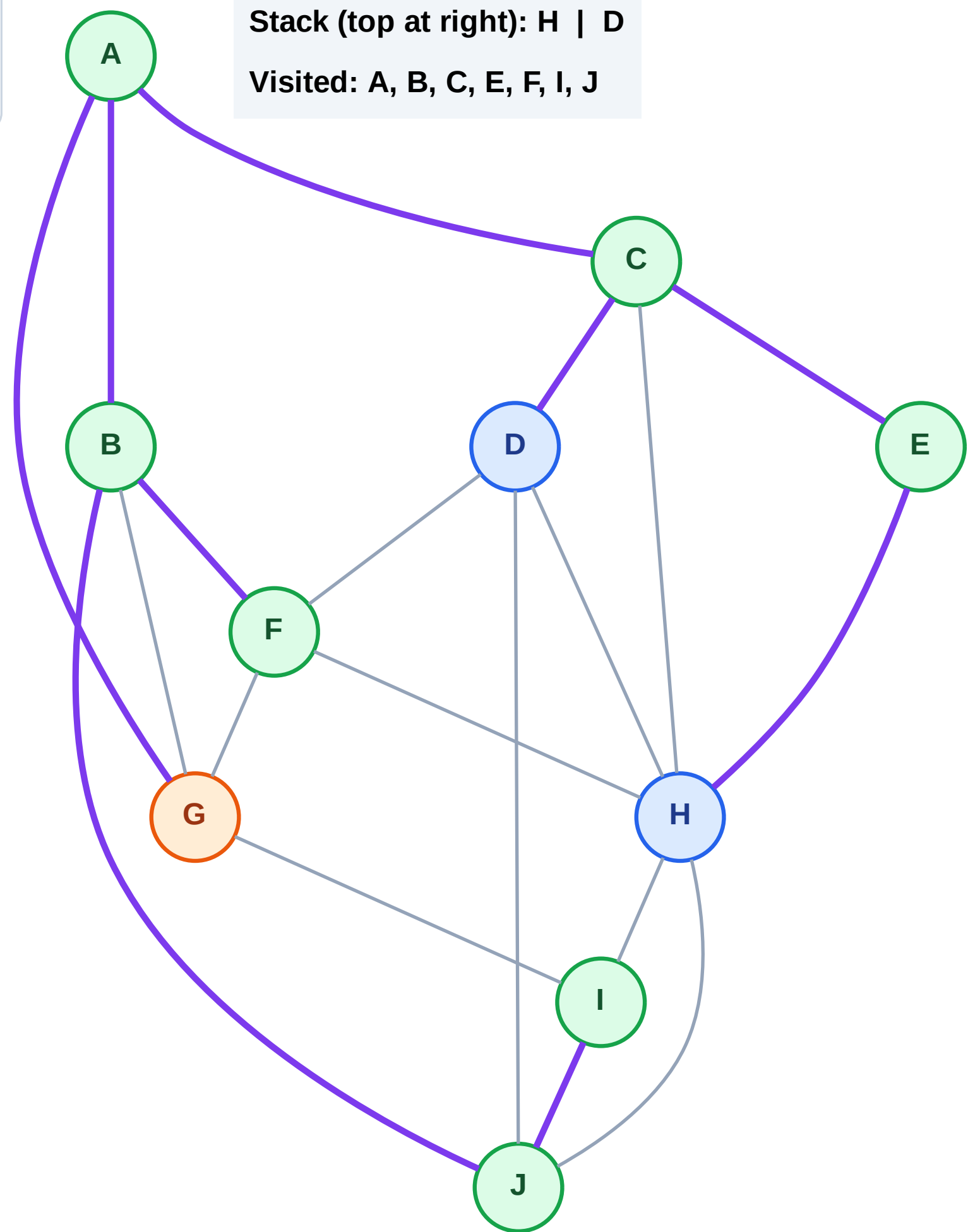
Step 16: Process node G

Legend



Stack (top at right): H | D

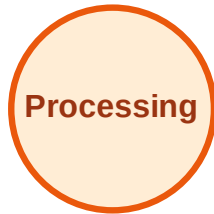
Visited: A, B, C, E, F, I, J



DFS Traversal

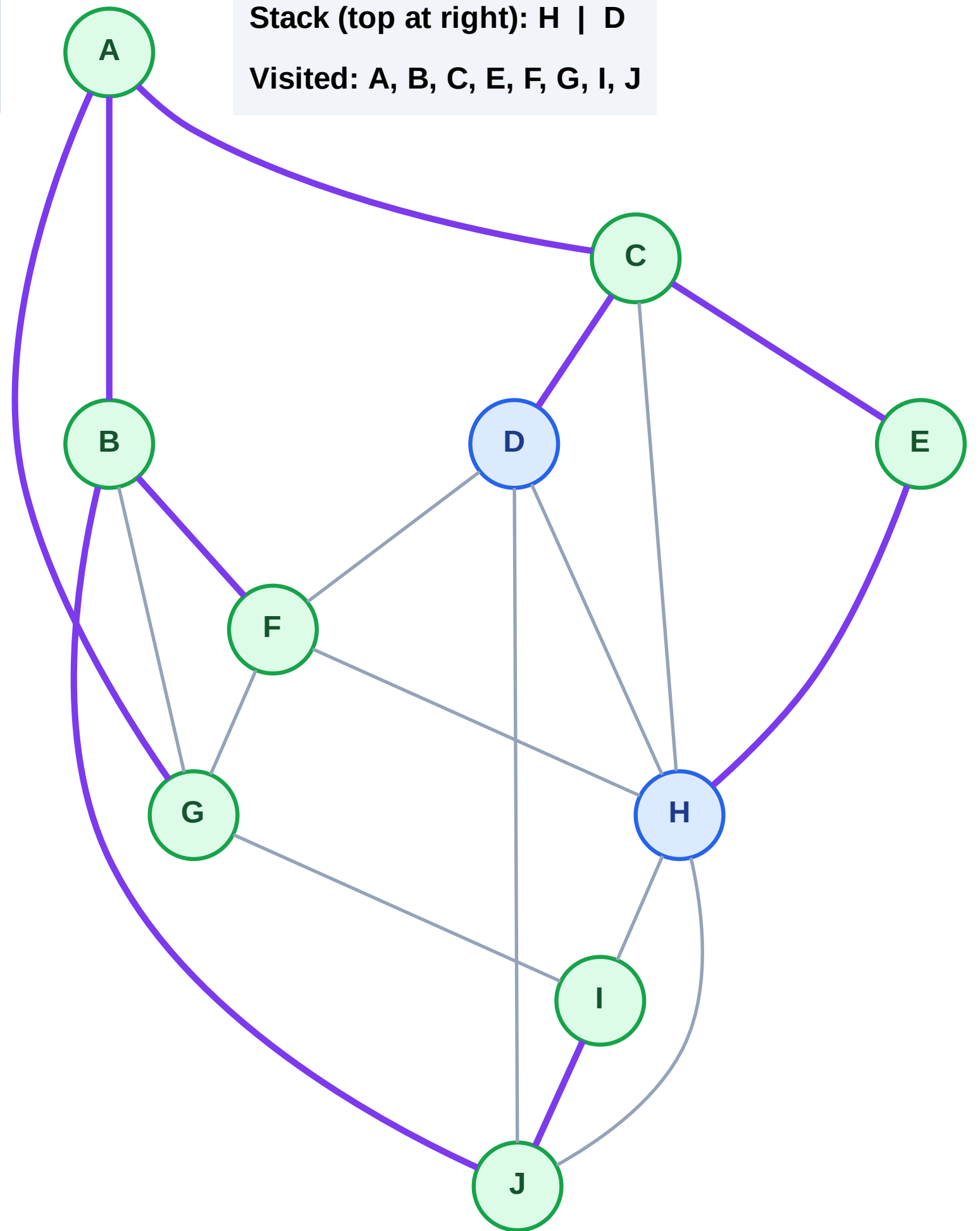
Step 17: No new neighbors from G

Legend



Stack (top at right): H | D

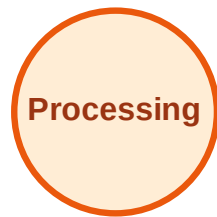
Visited: A, B, C, E, F, G, I, J



DFS Traversal

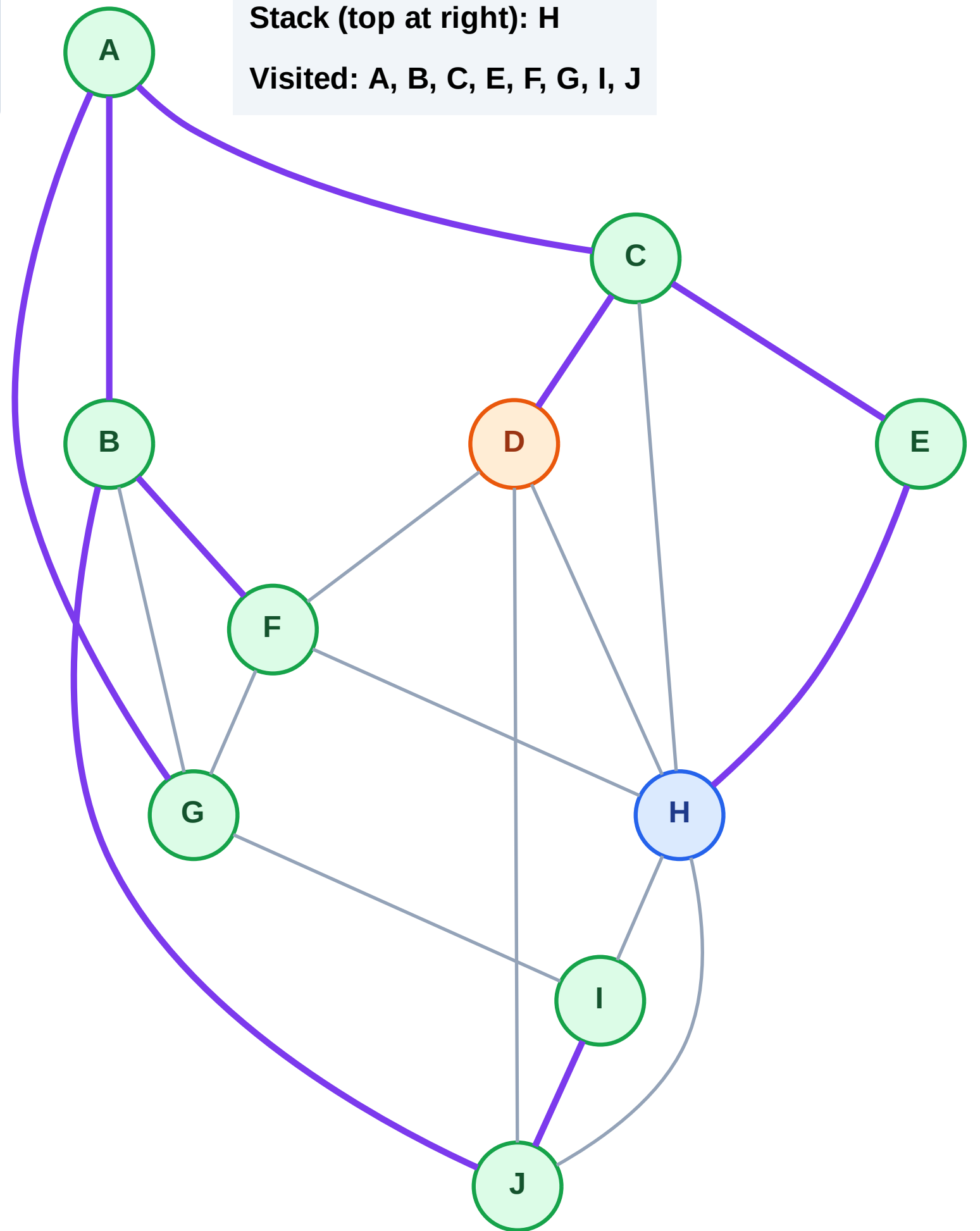
Step 18: Process node D

Legend



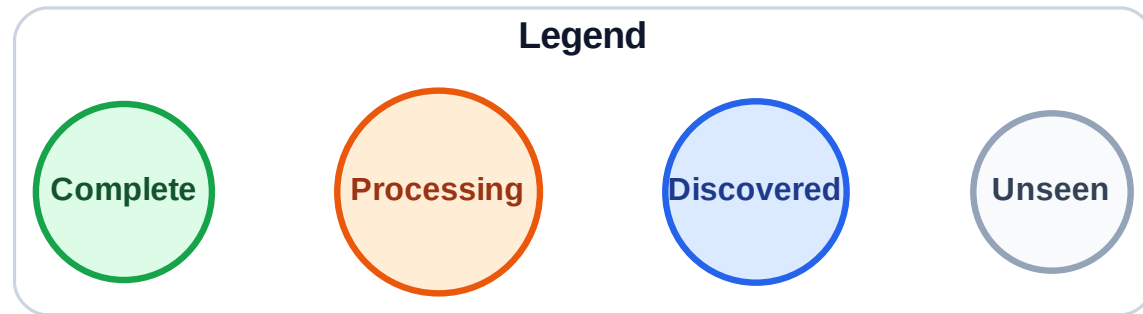
Stack (top at right): H

Visited: A, B, C, E, F, G, I, J



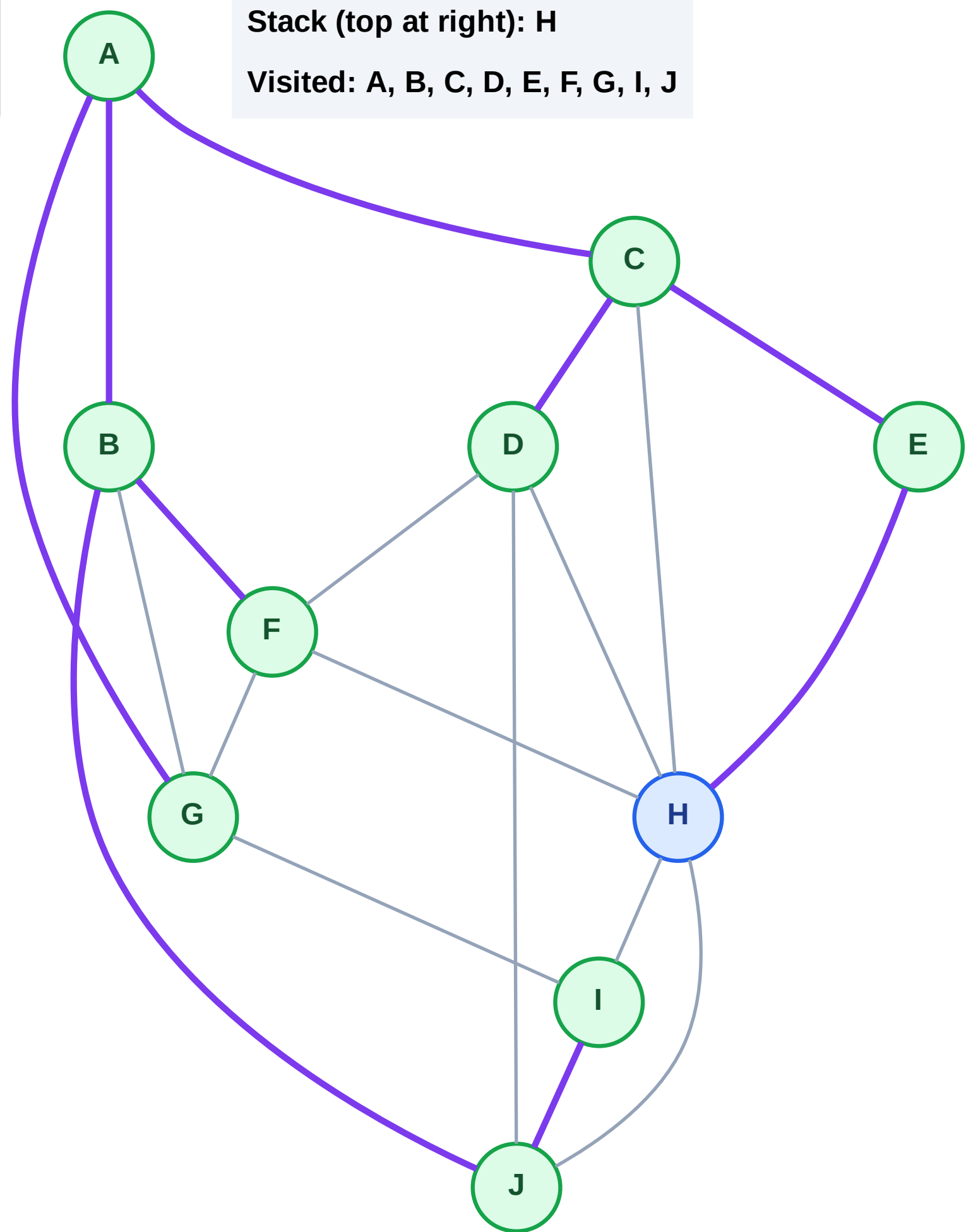
Step 19: No new neighbors from D

Step 19: No new neighbors from D



Stack (top at right): H

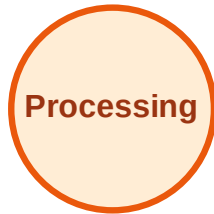
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

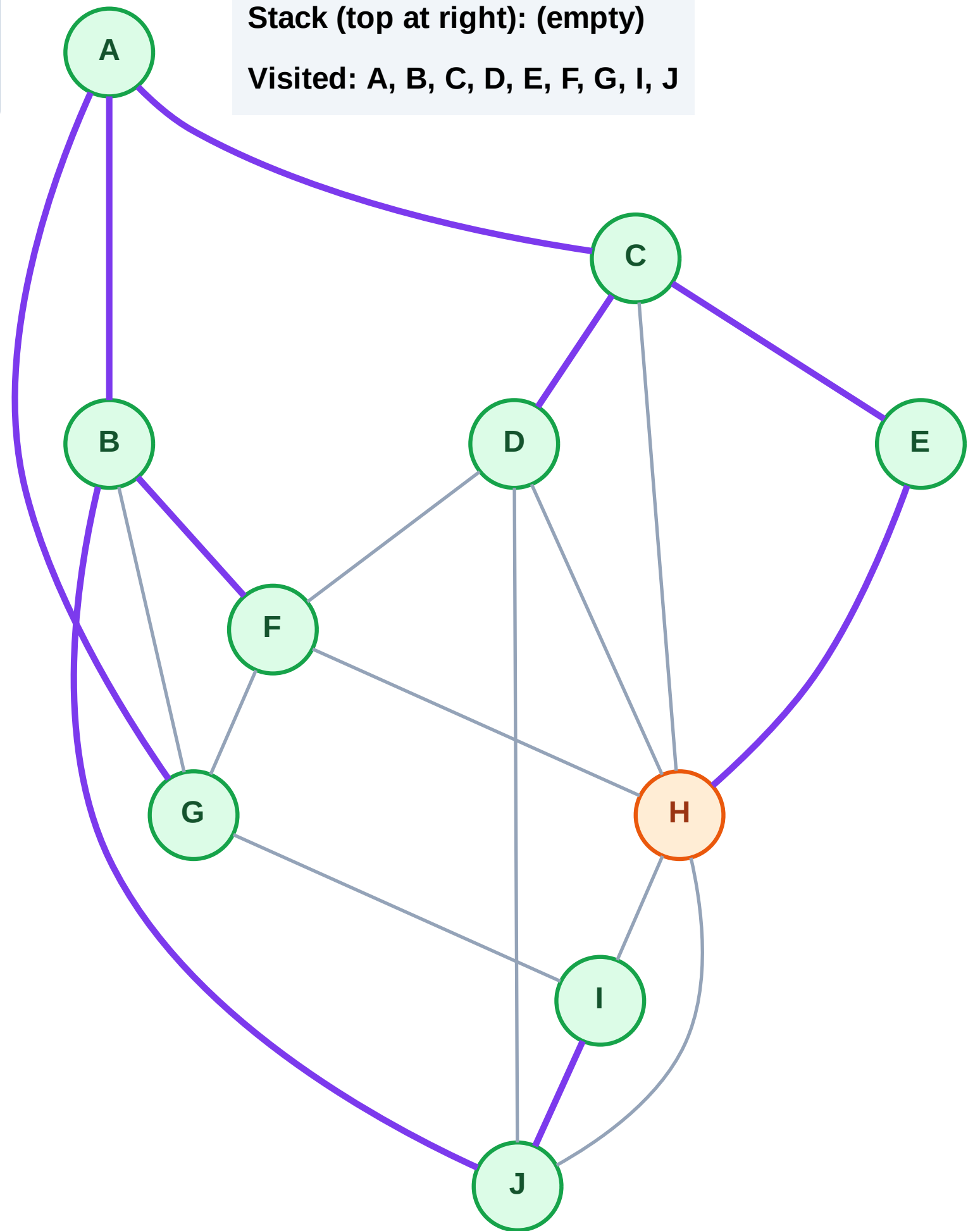
Step 20: Process node H

Legend



Stack (top at right): (empty)

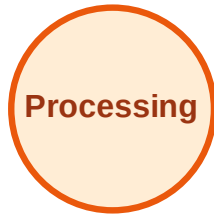
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

